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A Phylogenomic Investigation of Species Diversity, Morphology, and Biogeography in *Neptunia* (Fabaceae: Caesalpinioideae: Mimoseae)

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Abstract—*Neptunia* (Fabaceae) is a pantropical genus of 22 species and is the sole genus of Mimoseae with fully aquatic species, *N. oleracea* and *N. plena*. The aquatic species are pantropically distributed, while the 20 terrestrial species are confined within continents separated by oceanic barriers. Due to species having broad distributions, polyploidy, and cryptic morphology, *Neptunia* species delimitations remain uncertain and have not been comprehensively sampled in previous phylogenetic studies. To that end, we tested species delimitations in *Neptunia* by sampling multiple accessions of all species and infraspecific taxa and determined relationships through phylogenomics, identified the ancestral states of habitat and taxonomically defining characters, and estimated the historical biogeography of the group to determine the hypothetical ancestral area. A target-capture approach was applied with the Angiosperms353 bait-set to sequence 353 orthologous nuclear loci for phylogenetic inference of 94 ingroup and 11 outgroup individuals. Time calibrations and comparative methods were employed to reconstruct the chronogram and estimate ancestral states. Analyses of the filtered nuclear loci recovered a monophyletic *Neptunia* and provided insight into species delimitations in closely related diploid and polyploid species. We recognize *N. windleriana* and *N. heliophila* as synonyms based on phylogenetic reconstructions and morphology, reducing the number of species to 20. A newly described species, *N. proxima*, was polyphyletic. Single-seeded *Neptunia* species formed a clade, but other taxonomically important traits lacked phylogenetic signal. The aquatic *N. oleracea* was sister to the clade of the aquatic *N. plena* and the terrestrial species from the Americas, and ancestral character estimation analyses inferred two separate transitions into aquatic habitats. This transition into aquatic environments and being polyploid likely promoted the widespread distributions of the two aquatic *Neptunia* species. The early diversification of *Neptunia* was widespread in Asia, North America, or South America, and a rapid radiation occurred in Australia. Our results support recent taxonomic circumscriptions and provide the first fully sampled and well-resolved molecular phylogeny of *Neptunia*.

Keywords—Ancestral character estimations, Angiosperms353, aquatic species, historical biogeography, polyploid, target enrichment.

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Fabaceae includes an estimated 22,679 species and is an economically important group of crop and medicinal species (Lewis et al. 2005; LPWG 2025). Legumes are cosmopolitan, with different lineages being particularly species-rich in the tropics (e.g., *Mimosa* L., *Albizia* L., *Inga* L., *Parkia* R.Br.), arid or semitropical regions (e.g., *Dalea* L., *Acacia* L., and *Adesmia* DC.), and temperate regions (e.g., *Astragalus* L., *Lupinus* L., *Lotus* L., and *Trifolium* L.; Lewis et al. 2005; Ringelberg et al. 2023). Legume species have been successful in part due to their endosymbiotic relationships with atmospheric nitrogen-fixing bacteria, allowing species to colonize infertile areas and amend soils with ammonium and nitrate (Allen and Allen 1980). Legumes have evolved almost every growth form and occupy every biome with plant life (Lewis et al. 2005). While some species are widespread (e.g., *Medicago sativa* L., *Mimosa pudica* L.) others are endemic to narrow ranges (e.g., *Pictarena rotundata* [Wooton] Becklund; Becklund and Ayers 2022) or restricted within continents (e.g., *Dalea formosa* Torr.). The family is monophyletic and studies have found support for six main lineages that correspond to recognized subfamilies (Käss and Wink 1997; Doyle et al. 1997; Wojciechowski et al. 2004; Bruneau et al. 2008; LPWG 2017; Koenen et al. 2020c); however, efforts are still needed to understand species-level relationships and boundaries, particularly in species-rich or morphologically invariable genera with taxonomically ambiguous species or with broad distributions.

A legume genus that exemplifies ambiguous species boundaries is *Neptunia* Lour., a pantropical genus comprised of 22 species in the informally recognized *Dichrostachys* clade (Lewis and Elias 1981; Koenen et al. 2020a), in Mimoseae (formerly Mimosoideae; LPWG 2017; Hughes and Luckow 2024). *Neptunia* is named for Neptune, the Roman god of freshwater and oceans, in reference to the aquatic species *Neptunia oleracea* Lour. and *Neptunia plena* (L.) Benth. that occur in lakes, ponds, slow-moving parts of rivers, and ephemeral pools (Loureiro 1790; Cook 1990; Tabosa et al. 2012). The other 20 species of *Neptunia* are terrestrial and occur in marginal habitats, grasslands, tropical savannas, forest meadows, or open areas with full sun (Windler 1966; Silva et al. 2020; Bean 2022). The two aquatic species have pantropical distributions, while the terrestrial species occupy continental areas in Australia, Asia, and the Americas (Fig. 1).

Neptunia is characterized as a nodulating group of trailing herbaceous perennials and shrubs up to 1 m with leaflike stipules, thigmonastic bipinnate leaves, and heteromorphic inflorescences with sterile proximal flowers bearing yellow petaloid staminodes and distal perfect flowers with white filaments (Windler 1966; Allen and Allen 1980; Silva et al. 2020; Bean 2022). Species boundaries in *Neptunia* are defined by different configurations of several characters that are sometimes present or appear labile throughout the genus: an erect or prostrate habit, vestiture, the presence of petiolar nectaries, number of pinnae and leaflets, the presence of anther glands, and number of seeds per fruit (Fig. 2). A lack of staminodes in the proximal flowers is unusual in the genus but diagnostic in *Neptunia lutea* (Leavenw.) Benth., *Neptunia paucijuga* A.R.Bean, and *Neptunia proxima* A.R.Bean, although for the latter two species, the lack of staminodes could be attributed to available museum specimens (Bean 2022) or age (Windler 1966). *Neptunia* has a base chromosome number of $x = 14$ (Goldblatt 1981), and four species are known diploids: *Neptunia dimorphantha* Domin, *Neptunia monosperma* F.Muell. ex. Benth, *N. lutea*, and *Neptunia pubescens* Benth. (Turner and Fearing 1960). Three species are tetraploid ($2n = 56$),

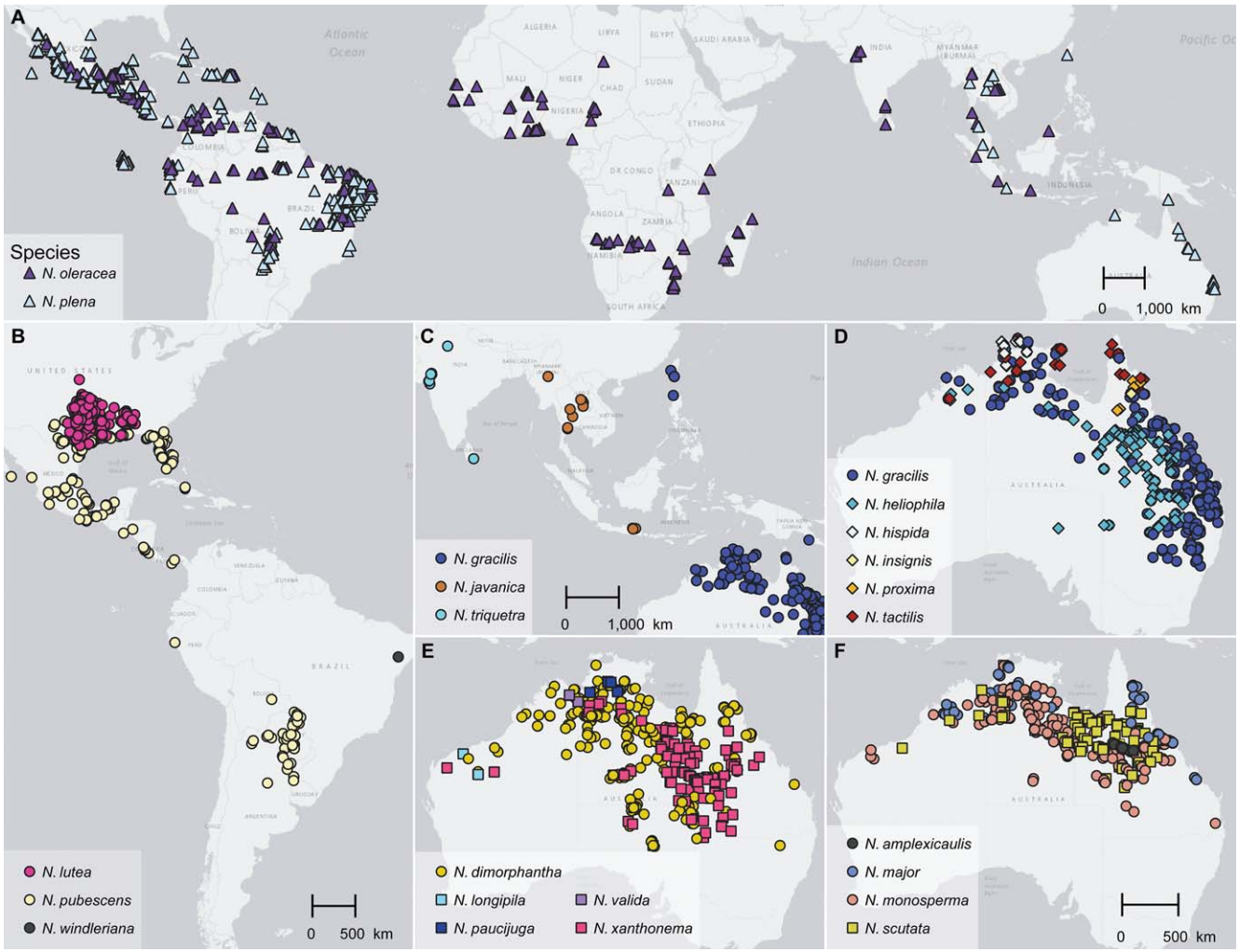

FIG. 1. Distributions of *Neptunia*. The aquatic species are triangles, the terrestrial species treated by Windler are circles, and the new species described by Bean from *N. gracilis* are diamond and from *N. dimorphantha* and *N. monosperma* (*N. scutata* only) are square. A. Pantropical aquatic species. B. Terrestrial species in North and South America. C. Terrestrial species of Asia. D–F. Terrestrial species of Australia. Species distributions are from herbarium records in digital repositories: Global Biodiversity Information Facility (GBIF 2024), Atlas of Living Australia (ALA 2024), Integrated Digitized Biocollections (iDigBio 2024), Naturalis Biodiversity Center (Naturalis 2024), and SEINet (2024).

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including *Neptunia gracilis* Benth. (Turner and Fearing 1960), *Neptunia windleriana* Santos-Silva & Mansano (Silva et al. 2020), and *N. oleracea* ($2n$ = [52, 54] 56; Miège 1960; Turner and Fearing 1960; Raman and Kesavan 1963; Sarkar et al. 1976; Kumar and Subramaniam 1986). A purported triploid *Neptunia triquetra* Benth. ($2n$ = 36) and hexaploid *N. plena* ($2n$ = 72 or 78) have a base chromosome number of x = 12 (Dnyansagar 1952; Frahm-Leliveld 1953; Berger et al. 1958; Turner and Fearing 1960; Pandit et al. 2006). The remaining 13 species have not been cytologically evaluated. The genus is notable for the aquatic crop species *N. oleracea* as a food vegetable (Csurhes 2016) and nitrogenous nutrient provider to rice crops (Hei et al. 2021, 2022) and, along with the terrestrial species *Neptunia amplexicaulis* Domin, known for hyperaccumulation of heavy metals (Atabaki et al. 2020; Harvey et al. 2020; van der Ent et al. 2023).

Until recently, *Neptunia* was comprised of 11 species and divided into two sections, *N.* section *Neptunia* Windler, with species that have decantherous flowers and occur in Asia, Africa, and the Americas, and *N.* section *Pentanthera* Windler with species that have pentantherous flowers and occur in Australia and the Philippines (Table 1; Windler 1966). Two recently described Australian species (Bean 2022) have stamen numbers that do not fit well into Windler's sections, with 5–8 stamens in *Neptunia tactilis* A.R.Bean and 9–10 stamens in *Neptunia hispida* A.R.Bean, and could represent a new section or indicate the need for revision in the sections of *Neptunia.*

From the earliest recorded accounts by Breyne (1678) to present, species boundaries in *Neptunia* have not been well understood due to cryptic morphology, the presence of putatively widespread species, underrepresentation in herbarium collections, and a lack of a robust phylogenetic hypothesis. Nomenclatural history provides additional confusion, as 15 species have heterotypic synonyms (Table S1; Turner 1951; Windler 1966; Bean 2022; IPNI 2025; POWO 2025). Variable character states in vestiture, pinnae number, leaflet number, and size of organs have confounded species boundaries. The treatment by Bean (2022) described 10 new species from five previously recognized species in Australia based on vestiture, pinnae number, petiolar nectary number, and leaflet size, and now include narrow endemics and broadly distributed

FIG. 2. Representative *Neptunia* species morphology: A–C. Habit of *N. plena*, *N. pubescens*, and *N. monosperma*. D–E. Heteromorphic inflorescences of *N. pubescens* and *N. gracilis*, with showy staminodes in sterile proximal flowers. F. Monomorphic inflorescence of *N. lutea*. G–H. Single-seeded fruit and petiole nectary of *N. monosperma*. I. Nectary-lacking petiole of *N. dimorphantha*. J. Anther gland of *N. pubescens*. K. Glandless anthers of *N. major*. L. Multi-seeded fruit of *N. pubescens*.

species that often co-occur with their source taxa (Fig. 1C–F). In the Americas, the broadly distributed *N. pubescens*, from the southeastern U. S. to central Argentina (Fig. 1), has several heterotypic synonyms based on variation in vestiture, pinnae number, and stipe length, which can vary over the growing season and damage from mowing (personal observation; Table S1). Most notably, the aquatic species *N. oleracea* and *N. plena* have been particularly taxonomically ambiguous for

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TABLE 1. The sections of *Neptunia* described by Windler (1966) and applied to the currently accepted taxa of *Neptunia*. Section *Neptunia* is defined by ten stamens; Section *Pentanthera* is defined by five stamens. Taxa with an asterisk (*) are species described after Windler's description.

Section *Neptunia* Windler (ten stamens)
**Neptunia hispida* A.R.Bean
Neptunia javanica Miq.
Neptunia lutea (Leavenw.) Benth.
Neptunia oleracea Lour.
Neptunia plena (L.) Benth.
Neptunia pubescens Benth.
Neptunia pubescens var. *microcarpa* (Rose) Windler
Neptunia triquetra Benth.
**Neptunia windleriana* Santos-Silva & Mansano
Section *Pentanthera* Windler (five stamens)
Neptunia amplexicaulis Domin
Neptunia amplexicaulis f. *richmondii* Windler
Neptunia dimorphantha Domin
Neptunia gracilis Benth.
Neptunia gracilis f. *glandulosa* Windler
**Neptunia heliophila* A.R.Bean
**Neptunia insignis* A.R.Bean
**Neptunia longipila* A.R.Bean
Neptunia major Windler
Neptunia monosperma F.Muell. ex Benth.
**Neptunia paucijuga* A.R.Bean
**Neptunia proxima* A.R.Bean
**Neptunia scutata* A.R.Bean
**Neptunia tactilis* A.R.Bean
**Neptunia valida* A.R.Bean
**Neptunia xanthonema* A.R.Bean

the same reasons. The two species are morphologically similar and separated by few characters: an unbranched versus branched stoloniferous habit (*N. oleracea* and *N. plena*, respectively), nectary-lacking versus nectary-bearing petioles, glandless versus gland-bearing anthers, and falcate versus oblong fruit (Windler 1966). Other characters, such as pinnae, leaflets, and flowers overlap in size and number (e.g., 30–50 flowers per head in *N. oleracea* and 30–60 in *N. plena*). The two aquatic species have overlapping ranges in the Americas and Asia (Fig. 1), with *N. oleracea* as the only species present in Africa. Although their native ranges are unknown before human influence, both species are considered invasive in Australia (Csurhes 2016) and *N. oleracea* is introduced in Arkansas (Peck and Serviss 2011), while a purportedly native population of *N. plena* is confined to Kenedy County, Texas (Richardson and King 2008). Several hypotheses on species boundaries in the aquatic species are equally plausible. The hypothesis of recognizing the two aquatic species was advocated by Windler in his monograph, but alternatively, recognition of a single aquatic species with variable morphology (Bean 2022), or as multiple cryptic species with continents separated by oceanic barriers are two alternative hypotheses. Furthermore, the morphological variation and ambiguity of *Neptunia* species, and particularly the aquatic species, could be due to recent hybridization, ancient hybridization, or effects from allopolyploidization. Molecular data collected from numerous populations across the species' geographic ranges and encompassing morphological variation is necessary to understand the species boundaries and interspecific relationships in *Neptunia*.

Previous studies have included a subsample of *Neptunia* species in phylogenetic investigations of the *Dichrostachys* clade and Mimoseae, and while *Neptunia* has been reconstructed as monophyletic, interspecific relationships were unresolved due to invariability and cytonuclear conflict among chloroplast and nuclear markers (Hughes et al. 2003; Luckow et al. 2003; Sulaiman et al. 2003; Koenen et al. 2020a; Ringelberg et al. 2022). A study of the nuclear ribosomal internal transcribed spacer (ITS) region included one individual for five species and two individuals for *N. plena* and *N. monosperma*, and recovered a polytomy of four separate lineages for Australia species, American species, *N. oleracea*, and *N. plena*, and a polyphyletic *N. monosperma* (Hughes et al. 2003). Other phylogenies based on the chloroplast markers *trnL* and *trnK* introns and spacer regions (Luckow et al. 2003) and *rbcL* (Sulaiman et al. 2003) included too few species of *Neptunia* to infer relationships, except that two accessions of *N. monosperma* were monophyletic (Luckow et al. 2003). The most recent molecular phylogenies of Mimoseae (Koenen et al. 2020a) and Caesalpinioideae (Ringelberg et al. 2022) included one species, *N. oleracea*, that was recovered in the *Dichrostachys* clade and sister to the Malagasy *Lemurodendron* Villiers &. P.Guinet. *Neptunia* has not been the focus of a molecular study and all species and infraspecific taxa with sampling across their ranges have not been phylogenetically evaluated.

Given the lack of resolution among species in the *Neptunia* clade, additional molecular variation is needed to resolve phylogenetic relationships. Target enrichment is a common reduced-representation phylogenomic approach that allows for sampling across the nuclear and organellar genomes in non-model organisms, has been effective at obtaining large datasets of hundreds of orthologs with variable and informative flanking intronic regions (Johnson et al. 2019; Hale et al. 2020; Larridon et al. 2020), and has potential in recovering well-supported clades in *Neptunia*. Target capture has also been successful in groups with polyploid species (e.g., Saunders et al. 2024; Tseng et al. 2024) with the aid of specialized filtering pipelines for ortholog assembly from target-capture reads (Yang and Smith 2014; Gardner et al. 2021; Morales-Briones et al. 2021; Nauheimer et al. 2021). Both lineage-specific (Kadlec et al. 2017; Shavvon et al. 2017; Vatanparast et al. 2018) and universal (Chau et al. 2018; Siniscalchi et al. 2021; Fabre et al. 2025) probe sets have been efficient at obtaining numerous orthologous loci in legumes and other systems. The lineage-specific Mimosoid bait set, "mimobaits," has successfully recovered 964 (Koenen et al. 2020a) and 997 (Ringelberg et al. 2022) orthologous loci for phylogenomic reconstruction of Mimoseae and Caesalpinioideae. While the combination of universal and lineage-specific baits is an ideal approach (Hendriks et al. 2021), the universal Angiosperms353 bait set has been effective across taxonomic levels of angiosperm groups, with added benefits of cost-effectiveness and the ability to combine data from species-level studies into higher taxonomic datasets, and is the choice bait set utilized for large-scale phylogenomic studies (e.g., PAFTOL, Genomics for Australian Plants; Baker et al. 2022; Bioplatforms Australia 2025). Among both bait sets, generic and species-level evolutionary relationships of legumes have been well supported and resolved recalcitrant phylogenetic issues resulting from repeated patterns of rapid diversification and polyploidization.

To delineate species boundaries and infer the evolutionary relationships of the 22 *Neptunia* species and infraspecific taxa, we sampled across their geographic ranges and morphological variation and utilized Angiosperms353 for phylogenomic evaluation. Our aims were to 1) clarify the inter- and

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intraspecific relationships within *Neptunia*, specifically the monophyly of the sections, the two widely distributed aquatic species, the newly described Australian species, and widely distributed *N. pubescens*; 2) determine if taxonomically important traits of the genus are orthologous (e.g., aquatic habitat, petiolar nectary, anther gland, number of seeds per fruit); and 3) understand the historical biogeography of the genus and determine if clades correspond with continental boundaries.

Materials and Methods

Sampling—To infer the evolutionary relationships among *Neptunia* taxa, DNA from leaflets of 120 individuals were sampled from silica-dried field collections or pressed herbarium specimens. Species in the United States were collected along roadsides in Texas, Arizona, and Florida and were deposited at BHO (abbreviations follow Index Herbariorum; Thiers 2025). Herbarium specimens were sampled from leaflet packets with permission from A, BRI, CANB, DNA, L, MO, TEX, US, and Z. At least one sample of the currently recognized 22 *Neptunia* species and the three infraspecific taxa were sampled, and multiple individuals of widespread species were sampled across their geographic ranges to assess species boundaries with monophyly. *Lemurodendron capuronii* Villiers & P.Guinet, *Dichrostachys cinerea* (L.) Wight & Arn., *Desmanthus velutinus* Scheele, *De. virgatus* (L.) Willd., and *Prosopidastrum mexicanum* (Dressler) Burkart were included as outgroups from closely related genera in the *Dichrostachys* clade (Hughes et al. 2003; Koenen et al. 2020a). *Calliandra eriophylla* Benth., *Acaciella angustissima* (Mill.) Britton & Rose, and the caesalpinoid *Parkinsonia aculeata* L. were included for fossil calibrations (see Chronogram below; Koenen et al. 2020b). Other outgroups included mimosoids *Neltuma glandulosa* (Torr.) Britton & Rose, *Albizia julibrissin* Durazz., and *Mimosa aculeaticarpa* Ortega. Previously sequenced accessions of *N. gracilis* and *Dichrostachys cinerea* were downloaded from GenBank and included in phylogenetic analyses (Appendix 1).

Target Enrichment and Sequencing—Genomic DNA from the 120 samples were extracted following the Schenk et al. (2023) CTAB v. 5 protocol based on Doyle and Doyle (1987). Due to polysaccharide and phenol coprecipitation, herbarium specimen extractions were further amended by adding 2% PVP-10 to the extraction buffer, a second chloroform isolation step, and precipitated for 2–4 days (Larridon et al. 2020). DNA concentrations were assessed with a Qubit fluorometer (Thermo Fisher Scientific, Waltham, Massachusetts, USA) and DNA degradation was assessed with 1% agarose gel electrophoresis.

Genomic library preparation was performed with the NEBNext Ultra II DNA Library Prep and NEBNext Ultra II FS DNA Library Prep kits (New England Biolabs, Ipswich, Massachusetts, USA) following the manufacturer's protocol. Approximately 100–500 ng of genomic DNA extract per species was enzymatically fragmented for 5–10 min; extracts shorter than 700 bp were not fragmented. After end repair and adaptor ligation, fragments were size selected for 200–350 bp inserts based on sample degradation and fragment size, following the NEB protocol with AMPure XP magnetic beads (Beckman Coulter, Brea, California, USA). Extracted DNA shorter than 300 bp were not size selected and instead cleaned with 0.8X AMPure XP magnetic beads during the size selection step of the NEB manufacturer's protocol. Samples were indexed with the 96 Index Primers NEB Multiplex Oligos for Illumina performed in 6–9 PCR cycles. DNA libraries were quantified using the Qubit fluorometer and fragment sizes were evaluated on the Agilent TapeStation (Agilent, Santa Clara, California, USA).

DNA libraries were enriched with the Angiosperms353 myBaits target capture bait set (Daicel Arbor Biosciences, Ann Arbor, Michigan, USA) following the myBaits Hybridization Capture Reaction for Targeted NGS protocol v. 4.00 for the first and v. 5.03 for the second and third sequencing runs. The Angiosperms353 bait-set was selected due to its use in broader legume molecular studies, such as PAFTOL, and its utilization in other studies in our lab (e.g., Fabre et al. 2025). Libraries were grouped into three pools (or a single pool for the MiSeq sequencing reaction) to normalize their concentrations for target enrichment reactions. The pools were lyophilized to 7 µl for the target enrichment reaction and were enriched in three separate reactions with the Angiosperms353 myBaits target-capture bait-set. The second and third sequencing rounds were enriched in half reactions of the baits with 2.75 µl of Ultrapure water to maintain the LIBs volume (Step S1.3 of the myBaits v. 5.03 protocol). Enrichment and bead wash were performed at 65°C. The final PCR step of the myBaits protocol was run for 16 cycles and the target enrichment pools were cleaned in a 0.9X AMPure XP bead cleanup. Target pool concentrations and fragment size range were quantified on the Agilent TapeStation. Due to fragments outside of the target range in one enriched pool in the second sequencing run, the pool was size selected with 40 µl followed by 20 µl of AMPure XP beads and re-quantified. In addition to the enriched libraries, an unenriched library "spike" was added to the enriched pools, amounting to 10% of the final pool concentration. For final pooling, we created a calculator to determine the volumes of each enriched and unenriched pool for a final pool comprised of 10% unenriched and 90% enriched libraries (available at https://github.com/lebecklund). To obtain longer reads and capture the variable intron regions adjacent to the targeted exons, sequencing was performed in three runs, two on the Illumina NovaSeq platform for 250 bp paired-end reads and one on the MiSeq for 300 bp paired-end reads, performed by the IGM Genomic Services Lab of the Abigail Wexner Research Institute at Nationwide Children's Hospital, Columbus, Ohio.

Sequencing Read Quality Control—After demultiplexing by the sequencing facility, individuals that were sequenced with two different indices were combined to increase read number. One sample (*Neptunia pubescens* LEB54) had > 300 M reads and was randomly subsampled to 15% of paired reads from one lane with SeqKit v. 2.7 using the sample function (sample –p 0.15 –s 11; Shen et al. 2016). Adaptors and low-quality sequences were removed with Trimmomatic v. 0.36 (Bolger et al. 2014), with adaptors trimmed with a seed length of two, a palindrome clip threshold of 30, a simple clip threshold of 10, a sliding window quality threshold of 4:15, a leading and trailing trimming of low-quality bases set to 3, and minimum sequence length of 57 or 130 (Table S2). Sample quality was evaluated with FastQC v. 0.12 (Andrews 2010). Cleaned, paired reads were used for downstream analyses.

Loci Assembly—To assemble the nuclear dataset targeted by the Angiosperms353 bait kit, sequences were compiled into the 353 genes with HybPiper v. 2.1.6 (Johnson et al. 2016) based on a gene target file that included all Fabaceae reference loci (8464 individuals, 567 taxa) from the 1000 Plants Project (One Thousand Plant Transcriptomes Initiative 2019). HybPiper is a python-based pipeline that utilizes BWA (Li and Durbin 2009) to map nucleotide sequences to the reference sequence files, separating sequences into their 353 nuclear loci. Genes were assembled using SPAdes (Bankevich et al. 2012) and aligned to the reference sequence with Exonerate (Slater and Birney 2005). The –run_intronerate flag was included in the assembly to capture flanking intron regions adjacent to the exons, and any contigs with < 10 read pools were removed. To obtain the phylogenetically informative intronic regions that have been effective in species-level phylogenetic analyses, we retrieved the "supercontig" sequences (exon and flanking intron regions) with retrieve_sequences.py.

Sample summary statistics were generated with the hybpiper_stats script in HybPiper (Johnson et al. 2016), which summarizes the number of genes that were assembled for each sample and the percentage of each gene that was recovered. If a sample had fewer than 100 genes that were > 25% recovered, it was removed from downstream analyses (Gardner et al. 2021), with two exceptions: *Neptunia gracilis* MR32898 and *N. lutea* SMT8023 were also removed due to their sequence quality and alignment issues (Appendix 1; Table S2), likely due to the age of the tissues (> 100 years old). Putative orthologous supercontig sequences were used in downstream analyses.

Gene Trees—For the four filtering steps up to the species tree analyses, we employed the same motif for alignments of each locus and gene tree reconstructions. Loci were aligned individually using MAFFT v. 7.520 (Katoh and Standley 2013) with 1000 maximum iterations and the Smith-Waterman algorithm for pairwise alignments (–localpair). Gene trees were reconstructed individually in RAxML v. 8 (Stamatakis 2014) under the GTR+Γ nucleotide substitution model and 1,000 maximum likelihood searches with rapid bootstrapping (–p 54321 –m GTRGAMMA –x 444 –# 1000 –f a).

Filtering Paralogous Loci—To obtain a dataset of orthologous loci for phylogenomic and comparative analyses, we identified and removed paralogous loci from the data. Gene paralogy, where two or more copies of the locus are found from target capture, can be caused by gene duplication, hybridization, or polyploidy. Five *Neptunia* species and our outgroup, *Lemurodendron capuronii*, are polyploids of unknown origin (Windler 1966; Silva et al. 2020; Ringelberg et al. 2022). In allopolyploid species, signatures of hybridization may be present, such as variable homoeologous genes of the different sub-genomes that have evolved independently and remain distinct enough to identify, while autopolyploid species might have fewer instances of paralogous loci due to a shared infraspecific history of the duplicated genomes. To account for

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genetic variation due to independent evolutionary histories of the polyploid sub-genomes, the assembled supercontigs underwent four filtering methods that identified paralogous loci and noise in the data set. In absence of a reference genome for *Neptunia*, paralogs were conservatively removed, rather than clustered or analyzed separately (Koenen et al. 2020a), to reduce the noise from artificially combining multiple copies of a locus among closely related species. First, in HybPiper (Johnson et al. 2016), the paralog_retriever python script identified paralogous sequences that were flagged as having contigs with deep read pools (> 10 times the adjacent contig) and if there were multiple versions of contigs that span at least 75% of the sequence. All paralogous loci were removed from downstream analyses.

In the second filtering step of supercontigs, we evaluated ambiguous nucleotide sites resulting from multiple alleles per site. Single nucleotide polymorphisms (SNPs) can be a common consequence of polyploidy and hybridization, and multiple SNPs from different genomes can result in conflicting gene tree topologies and increased phylogenetic noise. To detect loci with a high proportion of ambiguous sites, particularly from polyploids, HybPhaser v. 2.1 (Nauheimer et al. 2021) was used to reassemble loci and include ambiguity codes. The first step of the pipeline uses the HybPiper assembled supercontig (i.e., the consensus sequence) for each locus as the reference sequence to map sequencing reads in BWA (Li and Durbin 2009) and populate heterozygous sites with IUPAC ambiguity codes in SAMtools (Danecek et al. 2021). In the initial assembly, the –i flag was used to denote intronerated contigs. The number of ambiguous sites were then summarized in R v. 4.3.1 (R Development Core Team 2023). Outlier loci, or loci with SNPs above the third quartile of the total distribution of SNPs, were omitted from the HybPiper dataset. We used the HybPiper supercontig loci in downstream analyses for consistency, and removed aberrant samples detected with HybPhaser from the HybPiper locus files. Samples were removed if they had < 25% of the loci recovered and were < 10% of the target sequence length for a given locus. Entire loci were also removed if they had fewer than 30% of the samples, if < 25% of the locus was recovered, or if an outlier proportion of SNPs were recovered.

Another source of phylogenetic noise is individuals with a sequence that does not align, and results in an unusually long branch in the gene tree. In a third filtering step to identify tips with long branches, we utilized PhylteR v. 0.9.11 (Comte et al. 2023), which detects outlier tip-lengths to remove problematic accessions from alignments and does not manipulate the branch length of the gene tree. A PhylteR threshold of two was used to determine outlier tips in assembled gene trees. Flagged gene trees were inspected manually to verify the detected long branch. The outlier individuals were removed from the HybPiper supercontig files and a new alignment was generated in MAFFT under the aforementioned parameters. The gene trees were reconstructed in RAxML under the same parameters as outlined above.

As a final assessment of locus consistency, specifically with the species tree, the gene trees were evaluated for bipartition support in SortaDate (Smith et al. 2018) following Frost et al. (2024). Gene trees and the 105-tip species tree were rooted in Phyx v. 1.3.1 (Brown et al. 2017) with *Parkinsonia aculeata* assigned as the outgroup (–g). The gene trees were then compared to the species tree for bipartition support in SortaDate (Smith et al. 2018), and gene trees above the mean bipartition score were included in the most conservative species tree. Due to little difference in topology among the highest-filtered dataset and the dataset before the bipartition evaluation, the final dataset for the nuclear species tree consisted of 196 loci filtered in HybPiper, HybPhaser, and PhylteR.

Nuclear Species Tree—Phylogenetic inference was performed with the final, filtered dataset of gene trees (see Filtering Paralogous Loci below) with Astral III v. 5.7.5 (Zhang et al. 2018), a multispecies coalescent-based summary method that summarizes the relationships found in each gene tree and accounts for uncertainty of relationships with a local posterior probability (LPP) at each node. Two species trees were reconstructed from the gene trees with Astral III (Zhang et al. 2018) under default parameters. In the first Astral analysis, tips were not assigned to species due to the uncertainty in relationships among populations of widespread species. In a second Astral analysis, we assigned the multiple samples per species to designated species (–a), with exceptions in *N. windleriana* (assigned to *N. plena*), *N. gracilis* and *N. heliophila* (assigned to *N. gracilis* s.l.), and *N. proxima* (assigned to *N. proxima*-a and *N. proxima*-b). We included the three *N. gracilis* f. *glandulosa* individuals within *N. dimorphantha* following Bean (2022).

Chronogram—A chronogram was reconstructed in BEAST v. 2.7.7 (Bouckaert et al. 2019) for downstream comparative and biogeographic models. Due to the number of loci and individuals, our full dataset was too computationally intensive for BEAST, therefore a reduced dataset was utilized for the time-calibrated phylogeny. Informed by the species tree, the nuclear dataset was reduced to one representative per species that had the most loci with at least 75% coverage from HybPiper (Appendix 2). There were three exceptions to this criterion. First, both accessions of *N. proxima* were retained (as *N. proxima*-a and *N. proxima*-b) because the species was polyphyletic (see Results). Second, *Neptunia monosperma* DLL1068 was selected as representative for the species due to having more clocklike loci even though it had two fewer loci than other accessions. Third, because the *N. scutata* accession that best met our criterion was more closely related to another species (see Results), *Neptunia scutata* JE2995 was selected to represent *N. scutata*.

To further reduce our data to allow BEAST to run, the number of loci were subsampled based on three criteria: 1) loci that best matched the species tree topology, 2) loci with the lowest variance and followed the same clocklike pattern, and 3) loci that evolved under the same DNA substitution model. Reduction in the number of loci was necessary to achieve completed BEAST analyses in a reasonable time frame, to reach stationarity, and reach acceptable effective sample sizes (ESS) for the included parameters. We reduced the number of loci that were most clocklike and under the same substitution model as a means to determine the number of loci to include in BEAST analyses, rather than selecting an arbitrary number of loci. To test the first two criteria in SortaDate (Smith et al. 2018), the full dataset's (196) gene trees were rooted with Phyx (Brown et al. 2017) and the species tree was rooted in FigTree (http://tree.bio.ed.ac.uk/software/figtree). With SortaDate, we first selected loci based on the loci that were higher than the median quartet score (i.e., gene trees with highest number of nodes that matched the species tree; get_bp_genetrees.py) and second, the median number of loci that follow a clocklike pattern (i.e., lowest root-to-tip variance, get_var_length.py). The resulting loci were then reduced to their single species representative, aligned with MAFFT under the previous parameters, and concatenated with AMAS (Borowiec 2016). For our third criterion, the concatenated loci were then sorted by the their substitution models as determined with the –m TESTMERGE function in IQ-TREE 2 v. 2.3.5 (Minh et al. 2020). The substitution model with the greatest number of loci was selected for time-calibration analyses (GTR+I+Γ). The final 14 aligned loci were concatenated in AMAS (Borowiec 2016) under a single partition for time-calibration analyses.

Divergence time estimates were performed in BEAST with two fossil calibrations and were constrained to the species tree topology. Analyses were specified in Beauti v. 2.7.7 (Drummond et al. 2012). Analyses were run under the best-fitting GTR+I+Γ substitution model with starting values from the IQ-TREE 2 model output. An uncorrelated lognormal relaxed molecular clock was utilized with a pure-birth calibrated Yule model. Two fossil calibrations were included as gamma distribution priors. Former studies have expressed the importance of dating the root of the tree with a relevant fossil, including older fossils than represented by the node (Bell and Donoghue 2005; Schenk and Hufford 2010), and to avoid secondary calibrations to prevent artificially young estimates (Schenk 2016). A 58±1 Ma Eocene fossil of a bipinnately compound leaf calibrated the root of the phylogeny (Wing et al. 2009; Herrera et al. 2019), and while we did not sample the early-diverging lineages of Caesalpinioideae, the root occurred in the last 58 million years. A 16±1 Ma Miocene polyad pollen fossil of *Calliandra* Benth. (Caccavari and Barreda 2000) was placed at the node shared with the sister genus *Acaciella* Britton & Rose. *Acaciella* and *Calliandra* share eight-grain pollen polyads, but the fossil resembles *Calliandra* (Thulin 2023) and former the *Afrocalliandra* E.R.Souza & L.P.Queiroz species in shape (De Souza et al. 2013), and due to uncertainty of when the polyad evolved, we conservatively placed the calibration at the node. The analyses were constrained with the Astral III species tree topology that was first evaluated with penalized likelihood with the chronos function in the Ape R package (Paradis and Schliep 2019). To optimize branch lengths for time calibrations, the two fossil calibrations were included with a relaxed clock model (makeChronosCalib function) and the resulting tree was used as the BEAST multi-monophyletic constraint. A preliminary BEAST analysis was run for 31×10^6 generations, sampling every 3000 generations, to determine if the starting conditions under which the Markov-chain Monte Carlo (MCMC) performed were optimal. The log files were evaluated with Tracer v. 1.7.2 (Rambaut et al. 2018) and the updated starting prior and operator values were re-specified in Beauti. The final BEAST analyses were completed in two runs for 60×10^6 generations, sampling every 2000 generations. Stationarity, convergence, and parameter effective sample sizes for both runs were evaluated with Tracer (Rambaut et al. 2018) and the RWTY package in R (Warren et al. 2017) with a 20% burn-in. Trees from the two runs were

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merged with LogCombiner and final chronogram was consolidated in TreeAnnotator with 20% burn-in (Bouckaert et al. 2019) into the maximum clade credibility tree.

Hybridization of Nonnative Aquatics—*Neptunia plena* and *N. oleracea* are reportedly introduced in Australia based on the oldest herbarium records from 1960s, and are considered invasive in Queensland (Csurhes 2016). While the recent Australian treatment considered only *N. plena* as present in Australia (Bean 2022), *N. oleracea* is in cultivation in southeast Asia (Csurhes 2016). To understand their identity and detect the potential for hybridization among the two aquatics in their nonnative range, we tested for hybridization in the three Australian accessions among the two aquatic species with HyDe (phyde v. 1.0.2 Python module; Blischak et al. 2018). The 14 filtered alignments used for the species tree analysis were concatenated using AMAS (Borowiec 2016). A *N. oleracea* specimen from Thailand (KS36) and a *N. plena* specimen from Malaysia (RDW12337) were selected as the parental genomes to test for hybridization of *N. plena* JVWM198, *N. plena* LOsn, and *N. plena* JW3786. The morphologies of the specimens were also evaluated following Windler's treatment to confirm species identification.

Topology Hypothesis Test—Windler (1966) described two sections in *Neptunia*, *N.* section *Neptunia* with ten stamens and *N.* section *Pentanthera* with five stamens. To determine if the sections are phylogenetically supported, we performed an approximately unbiased (AU) test (Shimodaira 2002) on the 14 concatenated loci dataset in IQ-TREE 2 (Minh et al. 2020). The unconstrained tree and the constrained tree that follows the Windler sections (Table 1) were reconstructed with maximum likelihood for 10,000 bootstrap replicates under the GTR+I+Γ substitution model. The tree topologies were compared with the AU test (–au) for 10,000 bootstraps (–zb) under the GTR+I+Γ substitution model.

Ancestral States of Habitat and Traits—Two *Neptunia* species occur in aquatic ecosystems, however the ancestral ecosystem of the genus and the number of transitions from terrestrial to aquatic ecosystems in the genus is unknown. To determine the ancestral habitat, species were characterized as terrestrial or aquatic (Appendix 3) and evaluated with ancestral character estimation methods with the make.simmap function from the R package phytools v. 2.1.1 (Revell 2012) on the time-calibrated phylogeny. An equal rates model and an all-rates-different model were tested for the best-fitting state-transition-model with the fitDiscrete function in the Geiger package v. 2.0.11 (Pennell et al. 2014) and the best-fitting model for each trait was selected with the corrected Akaike information criterion (AICc). Ancestral character estimation analyses were run for 2×10^4 MCMC, sampled every 100 generations, and the variance of the Q value (vQ) was set to 0.01 (Revell and Harmon 2022).

Several morphological characters in *Neptunia* are diagnostic for species delimitations, many of which have quantitative or discrete states that overlap with other species, such as pinnae number, peduncle length, and vestiture. We do not know, however, if important diagnostic characters are homoplasious across the genus or follow infrageneric groups defined by Windler (1966). Definitive characters of *Neptunia* include the presence or absence of a petiole nectary, the presence or absence of an anther gland, and single-seeded or multi-seeded fruit. To evaluate the utility of those characters in classification of the genus, we conducted ancestral character estimations. Definitive characters were assigned to each species (Appendix 3): if a character was reportedly labile within a species, we coded the species as present for petiole nectary and anther gland, and the maximum number of seeds per fruit. The character states were modeled with ancestral state character reconstruction methods in make.simmap in phytools (Revell 2012) following the aforementioned conditions.

Historical Biogeography—The historical biogeographic patterns of *Neptunia* species were estimated with the chronogram in BioGeoBEARS v. 1.1.3 in R (Ree and Smith 2008; Matzke 2014) following the pipeline in Revell and Harmon (2022). We included all taxa in the time-calibrated phylogeny in analyses. A dispersal-extinction-cladogenesis (DEC) model was selected for the historical biogeographic inferences. We selected six regions following Ringelberg et al. (2023): North and Central America, South America, Africa, Madagascar, Asia, and Australia. The maximum number of areas that a species could occupy was set to five. Although the widespread *N. oleracea* and *N. plena* spanned multiple areas, they were constrained to a single region based on the phylogenetic results. We assigned the other species to their known native ranges ascertained from herbarium collections from the Global Biodiversity Information Facility (GBIF 2024), Atlas of Living Australia (ALA 2024), Integrated Digitized Biocollections (iDigBio 2024), Naturalis Biodiversity Center (Naturalis 2024), and SEINet (2024), personal collections, and literature (Windler 1966; Luckow 1993, 2023; Silva et al. 2020; Bean 2022; de Lourdes Rico-Arce 2023; Krings et al. 2023).

Results

Data—Raw sequence data are available on the NCBI Short Read Archive on GenBank, BioProject PRJNA1306678. A total of 120 individuals were sequenced. Fifteen individuals had too few or low-quality reads to assemble with HybPiper, most likely due to the condition or age of the specimens, including eight individuals of *Neptunia oleracea*, two individuals of *N. gracilis* outside of Australia, and the second accession of *N. triquetra*. The remaining 105 samples ranged from 217,210–47,298,854 (mean = 9,279,229) paired raw reads and 120,672–38,178,249 (mean = 8,018,596) paired cleaned reads per accession. The 363,714,688 raw reads for *N. pubescens* LEB54 were reduced to 13,621,500 paired raw reads for the pipeline. Loci that had at least 25% of the gene re-assembled with HybPiper ranged from 51–351 among the 105 samples.

Filtering and Final Datasets—Paralogous loci were detected in 145 loci with the paralog_retriever pipeline in HybPiper and were removed from downstream analyses. HybPhaser identified 10 paralogous loci and one locus with only three taxa, which we removed, resulting in 197 remaining loci. PhylteR identified 52 gene trees that had at least one outlier branch length; the outlier taxa were removed, the locus was realigned, and the gene tree was reconstructed. PhylteR identified one locus with 25 aberrant taxa, therefore the locus was removed from the final dataset and species tree. After the three paralog filtering steps, 196 loci remained and were summarized in the Astral species tree for additional filtering. After the fourth filtering step with SortaDate, 88 loci remained in the resulting species tree, the topology was the same as the 196-sample dataset but with lower node support. Therefore, we used the 196-loci dataset of 105 individuals to inform downstream analyses and taxonomic considerations. All sequence alignments are available on the Dryad Digital Repository (Becklund and Schenk 2025).

Nuclear Species Tree—The Astral species tree based on 196 loci reconstructed *Neptunia* as monophyletic and was sister to *Lemurodendron capuronii* with high support (LPP = 1.0; Fig. 3). *Neptunia* was comprised of two sister clades: a clade of the two pantropical aquatic species and America species, and a clade of the Asia and Australia species.

In the Pantropical and Americas clade (Fig. 3), the two pantropical aquatic species, *N. oleracea* and *N. plena*, were both non-monophyletic. *Neptunia oleracea* was polyphyletic with the inclusion of three hybridized *N. oleracea* × *N. plena* accessions that were embedded within the group (see Hybridization Analyses) from outside of the native ranges of *N. oleracea* and *N. plena*. A clade of *N. oleracea* accessions from Africa were sister to an Americas clade with high LPP, and the hybrids and Asia accessions were distantly related with moderate to low LPPs. *Neptunia plena* formed a paraphyletic grade of four lineages uncorrelated to geography or morphology. *Neptunia windleriana* was embedded within the *N. plena* clade and sister to a *N. plena* accession from Venezuela. One lineage of *N. plena* from South America was the closest relative to a clade of the two American terrestrial species, comprised of *N. pubescens* and *N. lutea*. The widespread terrestrial *N. pubescens* was monophyletic, but the infraspecific variety, *N. pubescens* var. *microcarpa*, was not monophyletic and did not have phylogenetic support as a variety. The accession of *N. lutea* var. *multipinnata* (BL2381; Turner 1951) was embedded within *N. lutea* var. *lutea* and did not have support as a separate variety.

In the Asia and Australia clade (Fig. 3), the three species that occur in Asia (*N. javanica*, *N. triquetra*, and *N. gracilis*)

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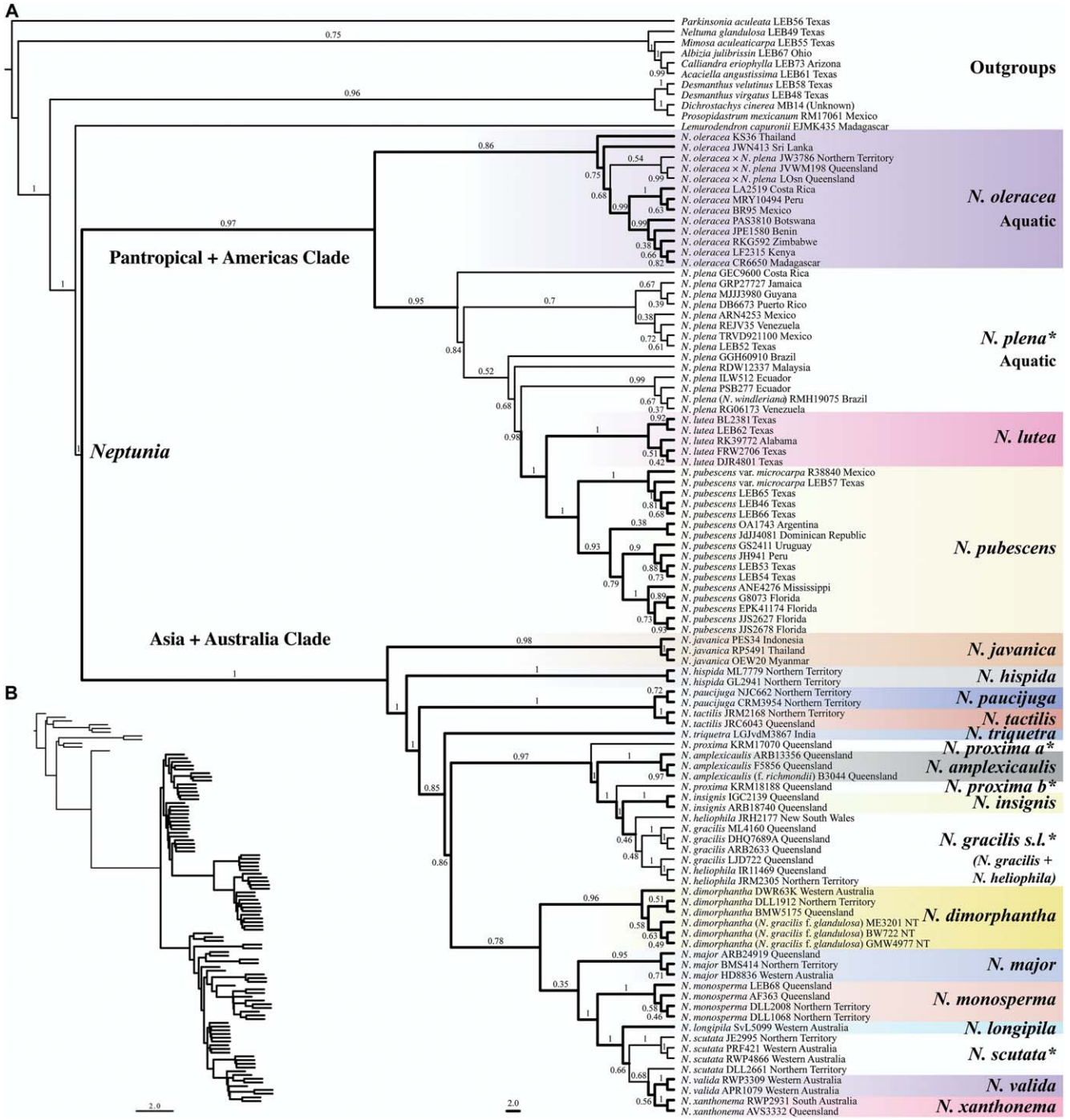

FIG. 3. The phylogeny of *Neptunia*. A. Cladogram of the Astral III species tree of 105 taxa summarized from 196 RAxML gene trees. Individuals are accompanied by their collector initials and number and their country origin (or state in countries with many collections). Individuals within a monophyletic species have bolded branches and are highlighted by colors corresponding to their symbol in Fig. 1. Non-monophyletic species are marked with an asterisk. Branch support values are local poster probabilities (LPP). B. The Astral phylogram with branch lengths. The ingroup is bolded. Tip branch lengths are an arbitrary length.

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did not form a clade together, and instead were sister to separate clades of Australian species. *Neptunia javanica* is the earliest branching species in the Asia and Australia clade and was sister to the remaining 16 species. *Neptunia triquetra* (LGJvdM3867), distributed from northeast India to Sri Lanka, was sister to a clade of 12 Australian species and was evidence of a second movement among Asia and Australia. Of the ten new Australian species, seven represented by more than one individual were monophyletic. *Neptunia proxima* was polyphyletic with one accession (KRM17070) related to a clade of five other Australian species, and the other individual within that clade (KRM18188) sister to *N. insignis*, *N. gracilis*, and *N. heliophila*. *Neptunia gracilis* and *N. heliophila* together formed a clade with accessions of the two species intermingled (Fig. 3), suggesting the two species are potentially conspecific in *N. gracilis* s.l. Finally, *N. scutata* was paraphyletic, with one accession (DLL2661) sister to the *N. valida* and *N. xanthonema* clade.

Chronogram—The chronogram (Fig. 4A) was reconstructed from 14 concatenated loci that all had the GTR+I+Γ DNA substitution model as the best-fit model. The two BEAST runs reached stationarity, converged on likelihood and posterior values, and effective sample sizes of parameter estimates exceeded 300 except for the UCLD standard deviation (ESS = 189) and the substitution rate for A to T (ESS = 171). Fossil-calibrated nodes (Fig. 4A) had broad estimates: 67.78 –25.89 Ma for the root node and 22.06–10.72 Ma for the *Calliandra* fossil, and mean ages resembled other time-calibrated legume studies (Lavin et al. 2005; Koenen et al. 2020b).

The *Neptunia* and *Lemurodendron* crown clade arose between 22.67–9.68 Ma (95% Highest Posterior Density [HPD]; Fig. 4A) and the *Neptunia* MRCA arose between 9.39–3.85 Ma. The Pantropic and Americas clade MRCA arose 9.04–3.65 Ma (Fig. 4A). The other major clade within *Neptunia*, comprised of ca. 16 species from Asia and Australia, arose between 5.82–2.42 Ma (crown node; Fig. 4A).

Hybridization Analyses—The hybridization hypotheses tested with HyDe revealed hybridization from parent species *Neptunia oleracea* and *N. plena* in *N. plena* JVWM198 ($p < 0.001$, Gamma = 0.44, Z-Score = 8.87) and *N. plena* JW3786 ($p < 0.001$, Gamma = 0.06, Z-Score = 37.82). Hybridization in *N. plena* LOsn was not detected ($p = 1$, Gamma = –0.02, Z-Score = -9.9×10^4).

Topology Hypothesis Test—In the comparison of the tree topology to Windler's sections, the tree that was constrained to Windler's sections had a significantly worse topology (–lnL = 297969.13, ΔlnL= 360.45, $p < 0.001$) than the unconstrained tree (–lnL = 297608.69). We determined that Windler's sections are not monophyletic and do not explain infrageneric clades within *Neptunia*.

Ancestral Character States—The equal-rate was the best-fitting model for petiole nectary, anther gland, seed number, and habitat (Table S3); it was applied in all ancestral character estimates. The ancestor of *Neptunia* was estimated to be terrestrial that independently transitioned into aquatic habitats twice (Fig. 4B). Among the ancestral states of petiole nectary, anther gland, and seed number, only seed number was a likely synapomorphic character for a clade of the single-seeded species (Fig. 5C) and the other traits were homoplasious across the genus (Figs. 5A, B).

Historical Biogeography of Neptunia—Assigned ranges for each species for historical biogeographic analysis are denoted at the branch tips (Fig. 6). The ancestral range of *Neptunia* was uncertain, with the highest proportion originating from either North America or Asia (Fig. 6) but also from Africa, Madagascar, or Australia. The ancestral range of the nodes along the backbone of the Asia and Australia clade were uncertain and were either derived from Australia or Asia. In a separate analysis, where *N. oleracea* was assigned to its widespread range in North America, South America, Africa, Madagascar, and Asia, and *N. plena* was assigned to North America, South America, and Asia, the ancestor of *Neptunia* was estimated to be widespread across the five continents in which *N. oleracea* occurs (Supplementary Fig. S1).

Discussion

We present the first fully sampled molecular phylogeny of *Neptunia*, a genus comprised of 20 species in the tropics and subtropics. *Neptunia* was comprised of two major clades, one clade of four pantropical and American species and second of ~16 species in Asia and Australia. The two clades did not correspond to Windler's sections (Windler 1966), which were rejected here. Windler's original *Neptunia* section *Pentanthera* with five stamens (*N. amplexicaulis*, *N. gracilis*, *N. major*, *N. dimorphantha*, and *N. monosperma*) was polyphyletic with

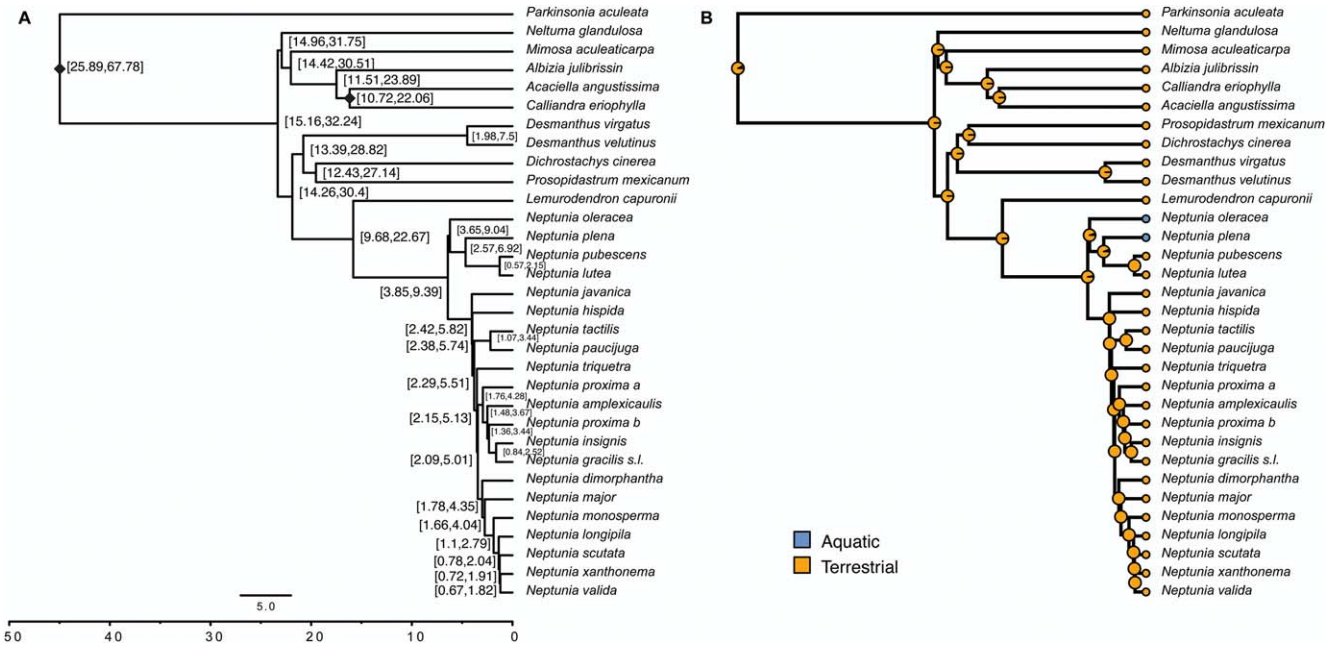

FIG. 4. *Neptunia* chronogram based on BEAST analyses. A. The chronogram with age ranges in millions of years (95% Highest Posterior Density [HPD]). The time-calibrated nodes from two fossil calibrations are denoted with diamonds: the root with a bipinnate leaf fossil from 59–57 Ma and the MRCA of *Calliandra* and *Acaciella* with a pollen polyad fossil from 17–15 Ma. B. Ancestral character state estimation for habitat in *Neptunia*. The tip colors show the habitat of the species, and the probabilities of the ancestral states for each node are given in pie charts.

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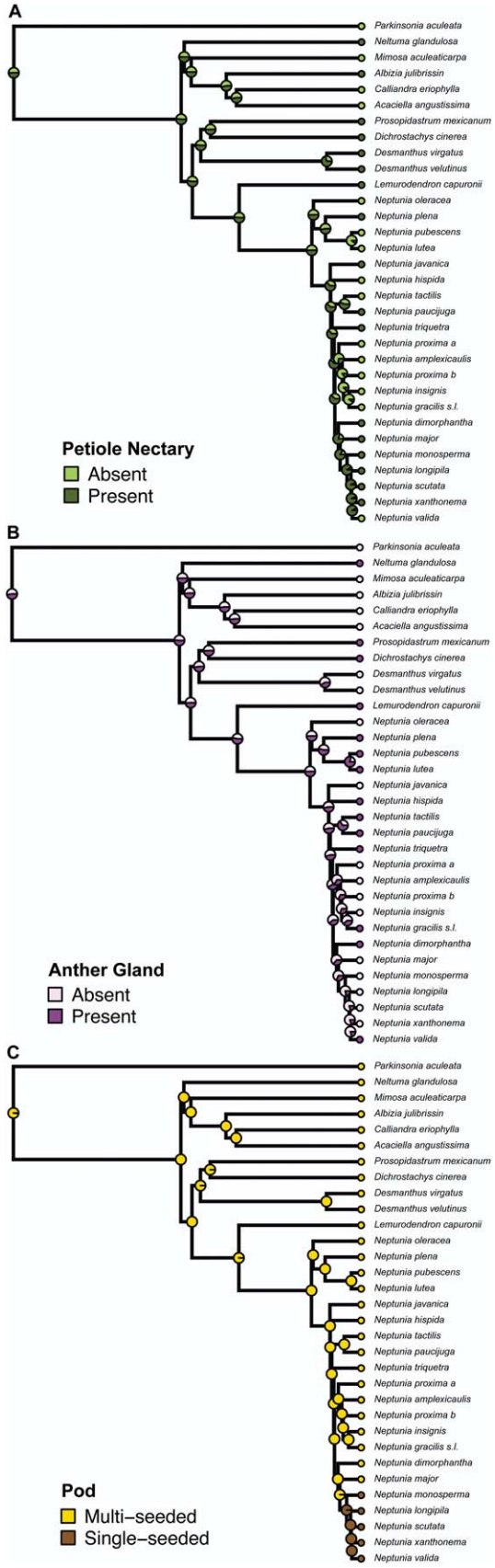

FIG. 5. Ancestral character state estimations of *Neptunia* for A. Petiole nectary. B. Anther gland. C. Single-seeded or multi-seeded fruit. Character states were assigned to each species, depicted at the tips. Character state uncertainty is shown in the pie charts for each state at each node.

the inclusion of the new Australian species, one of which, *N. paucijuga*, was sister to *N. tactilis* with 9–10 stamens (Fig. 3). *Neptunia* section *Neptunia*, with ten stamens (*N. oleracea*, *N. plena*, *N. pubescens*, *N. lutea*, *N. triquetra*, and *N. javanica*), was paraphyletic due to the close relationships of *N. triquetra* and *N. javanica* with species with five stamens (Fig. 3). Stamen number was homoplasious and precludes its use as a defining character for species clades in *Neptunia*, which seem to lack homologous traits and instead show geographic clustering.

A total of 196 orthologous loci were recovered with Angiosperms353 and a custom filtering pipeline, and most species relationships were resolved, including polyploids, with high support (Shah et al. 2021; Thomas et al. 2021; Ufimov et al. 2021). Angiosperms353 provided better phylogenetic resolution than traditional single-locus markers, and was as successful as customized bait-sets in recovering highly supported clades (Kadlec et al. 2017; Koenen et al. 2020a). Among the 105 accessions, 145 paralogous loci were identified with HybPiper, likely due in part to polyploidy; other studies of allopolyploid species found a high number of paralogous loci with HybPiper (e.g., up to > 80%; Morales-Briones et al. 2021). In absence of a reference genome, the origins of polyploidy in *Neptunia* species, and whether they are autopolyploid or allopolyploid, remain undetermined. Our phylogeny provides a framework for the further investigation into *Neptunia* cytology and the population-level studies necessary to disentangle their polyploidization.

Polyphyletic Terrestrial Species—Thirteen of the 22 *Neptunia* species sampled with multiple accessions were monophyletic, but seven were not monophyletic and merit review (the other two species were sampled from one accession). In the Pantropic and Americas clade, the recently described *N. windleriana* from Brazil (Silva et al. 2020) was embedded within *N. plena* as a diminutive terrestrial form outside of the current taxonomic delimitation. *Neptunia windleriana* differs from *N. plena* by having fewer flowers (8–18 versus 30–60), a shorter peduncle (1.5–2.5 cm versus 2.8–13 cm), a shorter stipe than the calyx (3–4 mm versus 3–9.1 mm), and was tetraploid versus hexaploid. *Neptunia plena* occurs in the narrow range of *N. windleriana* and is known to inhabit marginal (e.g., ephemerally wet) areas in addition to aquatic environments (Windler 1966), therefore, the diminutive morphology of *N. windleriana* could reflect a plastic response to the environment. We tentatively treat *N. windleriana* as a cytotype of *N. plena*.

Of the Asia and Australia clade, *Neptunia gracilis* and *N. heliophila* were paraphyletic and closely related (Fig. 3). Previously considered one species (*N. gracilis*; Windler 1966), *N. heliophila* differs from *N. gracilis* by having more pinnae (4–6 pairs versus 2–4 pairs), antrorse vestiture (versus glabrous or straight patent hairs), more flowers (25–40 versus 12–25), and an anther gland (versus lack of anther gland). *Neptunia gracilis* is tetraploid but the cytology of *N. heliophila* has not been evaluated. Both occur in Australia and have overlapping ranges (Fig. 1), with *N. heliophila* having a wider range across western, northern, and eastern Australia, and *N. gracilis* distributed across eastern Australia. A geographic gradient emerged in the phylogeny of *N. gracilis* s.l., with *N. heliophila* from New South Wales being sister to a *N. gracilis* clade from Queensland and the Northern Territory. *Neptunia gracilis* also occurs in Papua New Guinea, Indonesia, the

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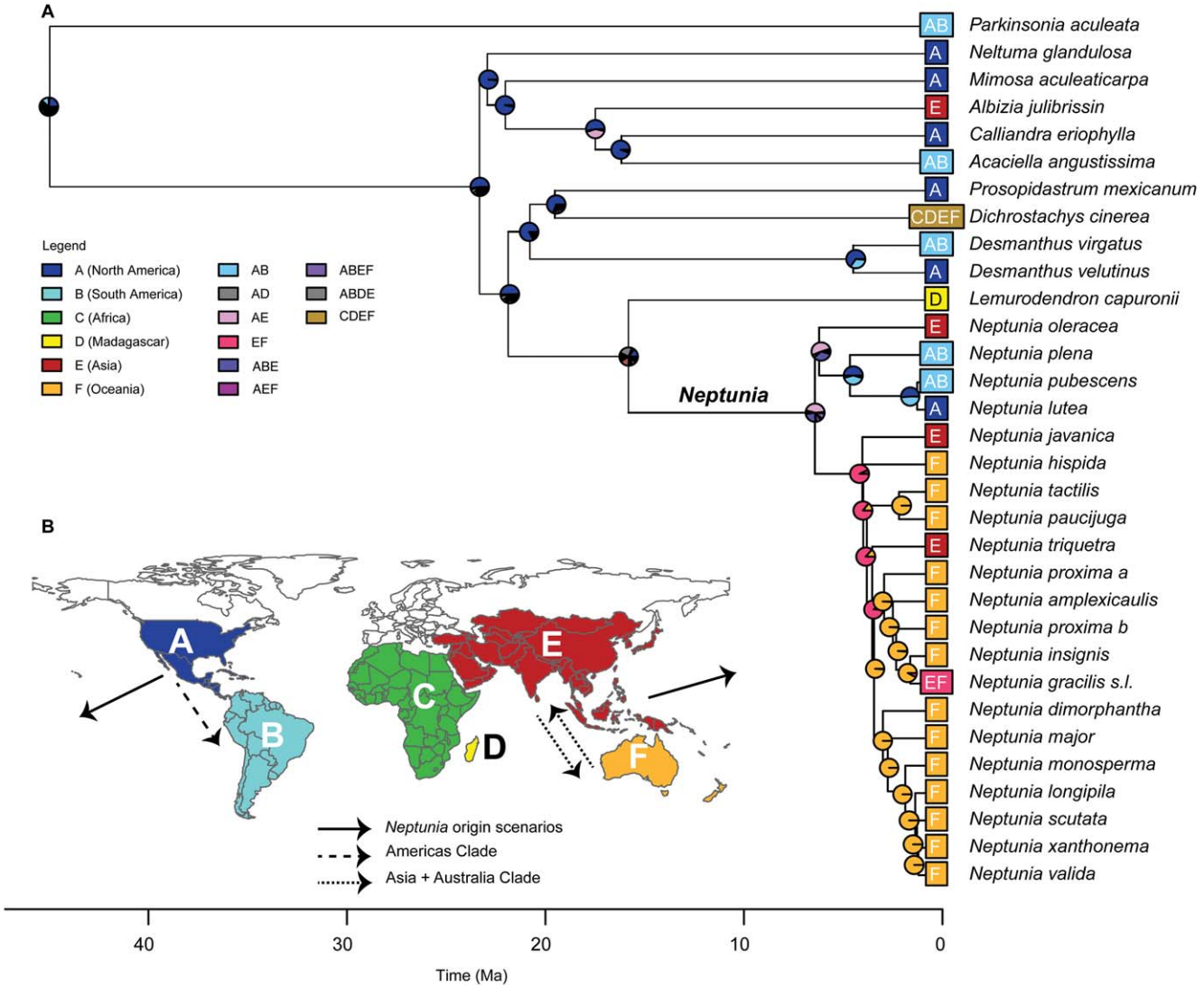

FIG. 6. The historical biogeographic estimations of *Neptunia* based on the DEC model. A. Species ranges were assigned at each tip. Ancestral range uncertainty is shown in the pie charts at each node; the hypothesized ancestral range of the root and deeper nodes are estimated, but due to lack of tips among Caesalpinioideae outside of *Neptunia*, were uncertain and appear black due to the pie chart borders of small proportions. B. The six geographic ranges assigned to the species. The arrows represent potential distribution patterns as determined in the BioGeoBears analysis.

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Philippines, and Taiwan but we were unable to amplify DNA of specimens from Papua New Guinea and the Philippines. Whether the two species are one polymorphic species, hybridizing, or related to due to effects of allopolyploidy are unknown and population-level sampling across their entire ranges is required to disentangle relationships.

Neptunia proxima, a newly described species in Australia, was polyphyletic (Fig. 3). The cause of the polyphyly is unclear; *Neptunia proxima* has a narrow range in northern Queensland (Fig. 1), a region where several Australian species co-occur (Bean 2022; personal observation). The two sampled accessions were included in Bean's (2022) treatment and varied from one another in the density of hairs and the presence of one bract on the peduncle of one specimen and lacking in the other. The species could represent two species with cryptic speciation, be a result of an allopolyploid event (ploidy of *N. proxima* is unknown), or one or both individuals could be influenced by hybridization; all scenarios are plausible as the species was previously circumscribed in the more-distantly related *N. gracilis*. Additional information is required to understand species delimitation of *N. proxima*.

Widespread Species—*Neptunia oleracea* and *N. plena* are the two aquatic species with pantropical (or nearly so) distributions and histories of ambiguous taxonomic boundaries. The phylogeny supports the current taxonomy of *N. oleracea* and *N. plena*, with morphologically similar accessions clustered with conspecifics. Both species have complex nomenclatural histories and have heterotypic synonyms based on the number of pinnae, number of leaflets, presence of aerenchymatous tissue, and geographic distribution. Their occasional occupation of terrestrial or marginal habitats have also confounded species delineations, as both can reportedly survive for at least part of their life cycle in seasonal floodplains and ephemeral pools (James et al. 2001; Silva et al. 2020), leading several botanists to consider terrestrial populations separate species (e.g., Windler 1966). Both species encompass a broad range in the number and size of morphological

characteristics, further evidenced by the current sampling of variable morphology. We hypothesized the two species could be one, two, or multiple cryptic species with geographic separation. If the two species were one species, as posited by Bean (2022), accessions with geographic proximity would likely be more closely related in the phylogeny or would have had closer evolutionary relationships than those that were recovered. Similarly, signatures of cryptic speciation, such as polyphyly or geographic structure, were absent. Phylogenetic relationships suggest both species are widespread and reached their current distributions before human influence (Fig. 3).

Neptunia oleracea was monophyletic with exclusion of the three hybrid accessions from Australia, and sampled individuals were grouped by continent. Individuals from the Americas were sister to those from Africa, and that clade is sister to the hybridized Australian populations, and a grade of Asian accessions were at the base of the species. Cultivated "water mimosa" populations were introduced to Australia in ~1960s as a food source from southeast Asia and *N. oleracea* and *N. plena* are considered invasive aquatic weeds throughout its range in Queensland and Northern Territory (Csurhes 2016). These purportedly introduced populations were included in the current study to understand the historical biogeography of the group, and our phylogeny supports their human introduction to Australia. Hybridization was detected in two of the three accessions from HyDe and intermediate morphological characteristics, supporting the hypothesis that the Australian accessions are cultivated hybrids, and would explain their inclusion within *N. oleracea*. While hybridization was not detected in one accession (LOsn; Fig. 3), the individual showed intermediate morphology among the two species and could be more closely related to the original populations of the hybrid accessions or an unsampled population, yielding a close relationship. The exact origins of the three Australian populations sampled here are unknown, and we consider them individuals that were hybridized from *N. oleracea* and *N. plena* due to their cultivation.

Neptunia plena was paraphyletic, with weak LPP support along the backbone and among the individuals of the four lineages, with the exception in the sister clade to the two terrestrial species, which had high LPP support. No clear morphological or geographic patterns were evident among accessions to merit infraspecific classifications. For example, *N. plena* REJ-V35 from Venezuela represented diminutive morphology similar to *N. windleriana*, but it was distantly related to the latter taxon, and both were more closely related accessions with aquatic morphology. *Neptunia plena* has a wide range of pinnae, leaflet number, and petiole and peduncle length, and can occupy ephemerally wet and aquatic environments from its northernmost boundary in southern Texas to central Argentina and into Asia (Windler 1966). In the evolution of paraphyletic species, as in *N. plena*, one population could have evolved into the ancestor of the two terrestrial species, which would indicate a reversion from an aquatic ancestor into a terrestrial environment and contradict our ancestral character estimations. Additionally, the potential for tetraploid and hexaploid populations could contribute to the uncertainty in the infraspecific relationships of *N. plena*. An expanded taxonomic delineation and population-level sampling of *N. plena* is required to understand the species.

Neptunia pubescens, the widespread terrestrial species in North and South America, was monophyletic in our phylogeny (Fig. 3). The infraspecific *N. pubescens* var. *microcarpa* was paraphyletic and does not merit an infraspecific rank. Of our sampled individuals, two well-supported clades were recovered, however, the clades are not morphologically distinct. One group is from Mexico and interior Texas, whereas the other group is from South America and gulf prairie in Texas, Mississippi, and Florida, adjacent to the Gulf of Mexico. The specimen from Central America did not sequence well enough for phylogenetic purposes. *Neptunia pubescens* was another species that has had taxonomic ambiguity and will need widespread collections and deeper population sampling to determine if there are infraspecific clades.

Character States—The ancestral character estimations revealed that most diagnostic traits were homoplasious and lacked phylogenetic signal, with one exception. Single-seeded pods were a synapomorphic character in an Australian *Neptunia* clade of five species, and likely evolved a single time (Fig. 5). Two closely related species were known to have one to three or five seeds, *N. dimorphantha* and *N. major*, indicating a reduction of seed number could have occurred in the clade. The other ancestral state character estimations determined petiolar nectary and anther gland as homoplasious traits across *Neptunia*, demonstrating their evolutionary complexity and repeated origins. Ninety percent of mimosoid genera have species with extrafloral nectaries, and in some cases are associated with ant-mediated defense (e.g., *Vachellia* Wight & Arn., *Inga*; Marazzi et al. 2019). Anther glands are also present among Mimosoid genera and are associated with pollination, either providing a pollinator reward, attractant, secreting sticky substances for pollen adherence, or protecting the anther during development (Luckow and Grimes 1997; de Barros and Teixeira 2016). *Neptunia pubescens* is the sole species of *Neptunia* that has been part of an anatomical study of anther glands, and bear a clavate anther gland that are thought to serve as attractants and aid in pollen dispersal during anther dehiscence (Tucker 1988; Luckow and Grimes 1997). While not phylogenetically associated, petiolar nectaries and anther glands may be correlated with other traits or environmental variables (Luckow and Grimes 1997) and need further investigation across Mimoseae.

Biogeographic Considerations—*Neptunia* arose 9.39–3.85 Ma during the late Miocene and Pliocene, with two main geographic clades splitting soon after. The ancestral range of *Neptunia* was not determined with certainty in the current study, with a proportion of the estimated origin from each geographic region but most likely originating from Asia or the Americas (Figs. 6, S1). The uncertainty in the two earliest nodes is likely related to the widespread distribution of *N. oleracea*. We included Madagascar as a separate geographic region due to the endemic sister species *Lemurodendron capuronii*, but the probability of a Madagascar origin was low. The ancestor of *N. plena*, *N. pubescens*, and *N. lutea* was estimated to originate in the Americas, which is where their ranges either overlap or are within 100 km (Fig. 1).

The clade of 16 species of Asia and Australia diversified more recently (5.82–2.42 Ma) and was more species-rich than the Pantropical and American species clade. Fourteen of the 16 species in the clade occur mainly in the northern half of Australia (Fig. 1) and the phylogeny suggests a relatively rapid radiation (Fig. 6). Two instances of the Asia species within a clade of Australian species suggests two dispersal events among the two continents, a surprising result in part due to the different section classifications among the Asia

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and Australia species; however, Bean (2022) remarked on the morphological similarities of *N. hispida* of the Northern Territory to *N. javanica*. The species *N. javanica* crosses Wallace's line, and its relationship to the Australian species suggest at least one crossing of Lyddecker's line (Rowe et al. 2019). The relationships of *N. triquetra* and *N. gracilis* to the other Australia species suggest two additional crossings of Wallace's line. We do not know if species moved from west to east as found in other studies on legumes and murine rodents (Sirichamorn et al. 2014; Rowe et al. 2019), or in the less common pattern from east to west, but either are plausible. The current understanding of *Neptunia* species ranges in Asia could be improved, as they are represented by few digitized museum collections.

Neptunia diversified in a similar time period to other globally distributed legume genera, such as *Calliandra* and *Mimosa* (Simon et al. 2011; de Souza et al. 2013), and transcontinental angiosperm clades (Lavin et al. 2004), supporting a long-distance dispersal theory rather than vicariance of continents. Transcontinental dispersal across oceanic barriers were as common as phylogenetic niche turnover in the evolutionary history of Mimoseae (Ringelberg et al. 2023), and the propensity of the group to inhabit aquatic and marginal habitats would aid in both proximity to oceanic waterways and survival of inundation. The dispersal mechanisms of *Neptunia* species remain unknown. Aquatic *Neptunia* could have vegetatively fragmented and floated on currents to reach pantropical distributions, but this has not been tested. In other genera with aquatic legumes, young plants, fruits, and seeds are known to float (e.g., *Aeschynomene*, *Sesbania*; Tyndall 2011; Farruggia et al. 2018) however these traits have not been examined in *Neptunia*.

Most of Fabaceae is terrestrial, with an estimated 150 species occurring in aquatic or semiaquatic environments (Cook 1990). Ancestral character state estimations support a terrestrial ancestor, with two separate transitions into aquatic species (Fig. 4). The aquatic *Neptunia* species form aerenchyma in the root cortex (Seago et al. 2005) and stem phellem (Metcalfe 1931), presumably to aerate tissues and float in the upper part of the water column. The aquatic *Neptunia* are notable because of their economic uses, but the other genera with aquatic species are distantly related and classified in the Papilionoideae, including *Aeschynomene* L., *Sesbania* Adans., and *Discolobium* Benth. (Cook 1990). Both *Sesbania* and *Aeschynomene* are pantropical genera (Brottier et al. 2018; Farruggia et al. 2018; Cardoso et al. 2020). Adapting to aquatic environments, as well as being polyploid, are likely related to the broad geographic distributions in the aquatic *Neptunia* species.

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Author Contributions

LEB contributed to project design, data collection, laboratory work, analyses, and the writing of the manuscript. JJS contributed to project design, data collection, guidance in analyses, and reviews of the manuscript.

Literature Cited

ALA. 2024. Atlas of Living Australia. https://doi.org/10.26197/ala.be498108-93d9-42bd-9fbd-64e150d07eaa (last accessed 19 February 2024).

Allen, O. N. and E. K. Allen. 1980. *The Leguminosae: A Source of Characteristics, Uses and Nodulation.* Madison: The University of Wisconsin Press.

Andrews, S. 2010. FASTQC: A quality control tool for high throughput sequence data. http://www.bioinformatics.babraham.ac.uk/projects/fastqc/.

Atabaki, N., N. A. Shaharuddin, S. A. Ahmad, R. Nulit, and R. Abiri. 2020. Assessment of water mimosa (*Neptunia oleracea* Lour.) morphological, physiological, and removal efficiency for phytoremediation of arsenic-polluted water. *Plants* 9: 1500.

Baker, W. J., P. Bailey, V. Barber, A. Barker, S. Bellot, D. Bishop, L. R. Botigué, G. Brewer, T. Carruthers, J. J. Clarkson, J. Cook, R. S. Cowan, S. Dodsworth, N. Epitawalage, E. Françoso, B. Gallego, M. G. Johnson, J. T. Kim, K. Leempoel, O. Maurin, C. Mcginnie, L. Pokorny, S. Roy, M. Stone, E. Toledo, N. J. Wickett, A. R. Zuntini, W. L. Eiserhardt, P. J. Kersey, I. J. Leitch, and F. Forest. 2022. A comprehensive phylogenomic platform for exploring the angiosperm tree of life. *Systematic Biology* 71: 301–319.

Bankevich, A., S. Nurk, D. Antipov, A. A. Gurevich, M. Dvorkin, A. S. Kulikov, V. M. Lesin, S. I. Nikolenko, S. Pham, A. D. Prjibelski, A. V. Pyshkin, A. V. Sirotkin, N. Vyahhi, G. Tesler, M. A. Alekseyev, and P. A. Pevzner. 2012. SPAdes: A new genome assembly algorithm and its applications to single-cell sequencing. *Journal of Computational Biology* 19: 455.

Bean, A. R. 2022. A revision of *Neptunia* Lour. (Leguminosae: Subfamily Caesalpinioideae, Mimosoid clade) in Australia and Malesia. *Austrobaileya* 12: 59–106.

Becklund, L. E. and T. J. Ayers. 2022. The Phylogeny of *Errazurizia* (Fabaceae: Amorpheae) and description of the new monotypic genus *Pictarena*. *Systematic Botany* 47: 1012–1024.

Becklund, L. E. and J. J. Schenk. 2025. Data from: A phylogenomic investigation of species diversity, morphology, and biogeography in *Neptunia* (Fabaceae: Caesalpinioideae: Mimoseae). Dryad Digital Repository. https://doi.org/10.5061/dryad.9kd51c5xg.

Bell, C. D. and M. J. Donoghue. 2005. Dating the Dipsacales: Comparing models, genes, and evolutionary implications. *American Journal of Botany* 92: 284–296.

Berger, C. A., E. R. Witkus, and R. M. McMahon. 1958. Cytotaxonomic studies in the Leguminosae. *Bulletin of the Torrey Botanical Club* 85: 405–415.

Bioplatforms Australia. 2025. Genomics for Australian Plants. https://www.genomicsforaustralianplants.com/ (last accessed June 2025).

Black, J. M. 1948. *Flora of South Australia. Part II.* Adelaide: Government Printer.

Blischak, P. D., J. Chifman, A. D. Wolfe, and L. S. Kubatko. 2018. HyDe: A python package for genome-scale hybridization detection. *Systematic Biology* 67: 821–829.

Bolger, A. M., M. Lohse, and B. Usadel. 2014. Trimmomatic: A flexible trimmer for Illumina sequence data. *Bioinformatics* 30: 2114–2120.

Borowiec, M. L. 2016. AMAS: A fast tool for alignment manipulation and computing of summary statistics. *PeerJ* 4: e1660.

Bouckaert, R., T. G. Vaughan, J. Barido-Sottani, S. Duchêne, M. Fourment, A. Gavryushkina, J. Heled, G. Jones, D. Kühnert, N. de Maio, M. Matschiner, F. K. Mendes, N. F. Müller, H. A. Ogilvie, L. Du Plessis, A. Popinga, A. Rambaut, D. Rasmussen, I. Siveroni, M. A. Suchard, C.-H. Wu, D. Xie, C. Zhang, T. Stadler, and A. J. Drummond. 2019. BEAST 2.5: An advanced software platform for Bayesian evolutionary analysis. *PLOS Computational Biology* 15: e1006650.

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Breyne, J. 1678. *Exoticarum Aliarumque Minus Cognitarum Plantarum Centuria Prima*. Gdańsk: David-Fridericus Rhetius.

Brottier, L., C. Chaintreuil, P. Simion, C. Scornavacca, R. Rivallan, P. Mournet, L. Moulin, G. P. Lewis, J. Fardoux, S. C. Brown, M. Gomez-Pacheco, M. Bourges, C. Hervouet, M. Gueye, R. Duponnois, H. Ramanankierana, H. Randriambanona, H. Vandrot, M. Zabeleta, M. DasGupta, A. D'Hont, E. Giraud, and J.-F. Arrighi. 2018. A phylogenetic framework of the legume genus *Aeschynomene* for comparative genetic analysis of the nod-dependent and nod-independent symbioses. *BMC Plant Biology* 18: 333.

Brown, J. W., J. F. Walker, and S. A. Smith. 2017. Phyx: Phylogenetic tools for unix. *Bioinformatics* 33: 1886–1888.

Bruneau, A., M. Mercure, G. P. Lewis, and P. S. Herendeen. 2008. Phylogenetic patterns and diversification in the caesalpinioid legumes. *Botany* 86: 697–718.

Caccavari, M. and V. Barreda. 2000. A new calymmate mimosoid polyad from the Miocene of Argentina. *Review of Palaeobotany and Palynology* 109: 197–203.

Cardoso, D., C. Mattos, F. Filardi, A. Delgado-Salinas, M. Lavin, P. Moraes, F. Tapia-Pastrana, and H. Lima. 2020. A molecular phylogeny of the pantropical papilionoid legume *Aeschynomene* supports reinstating the ecologically and morphologically coherent genus *Ctenodon*. *Neodiversity* 13: 1–38.

Chau, J. H., W. A. Rahfeldt, and R. G. Olmstead. 2018. Comparison of taxon-specific versus general locus sets for targeted sequence capture in plant phylogenomics. *Applications in Plant Sciences* 6: e1032.

Comte, A., T. Tricou, E. Tannier, J. Joseph, A. Siberchicot, S. Penel, R. Allio, F. Delsuc, S. Dray, and D. M. de Vienne. 2023. PhylteR: Efficient identification of outlier sequences in phylogenomic datasets. *Molecular Biology and Evolution* 40: msad234

Cook, C. D. K. 1990. *Aquatic Plant Book*. The Hague: SPB Academic Publishing.

Csurhes, S. 2016. *Invasive Plant Risk Assessment: Water mimosa* Neptunia oleracea *Dead and Awake* Neptunia plena. State of Queensland: Department of Agriculture and Fisheries.

Danecek, P., J. K. Bonfield, J. Liddle, J. Marshall, V. Ohan, M. O. Pollard, A. Whitwham, T. Keane, S. A. McCarthy, R. M. Davies, and H. Li. 2021. Twelve years of SAMtools and BCFtools. *GigaScience* 10: giab008.

De Barros, T. C. and S. P. Teixeira. 2016. Revisited anatomy of anther glands in mimosoids (Leguminosae). *International Journal of Plant Sciences* 177: 18–33.

De Lourdes Rico-Arce, M. 2023. *Albizia*. Pp. 136–138 in *Flora of North America North of Mexico*, vol. 11, Part 1, eds. Editorial Committee. New York and Oxford: Flora of North America Editorial Committee.

De Souza, É. R., G. P. Lewis, F. Forest, A. S. Schnadelbach, C. van den Berg, and L. P. de Queiroz. 2013. Phylogeny of *Calliandra* (Leguminosae: Mimosoideae) based on nuclear and plastid molecular markers. *Taxon* 62: 1200–1219.

Dnyansagar, V. R. 1952. Embryological studies in the Leguminosae IV: A contribution to the embryology of *Neptunia triquetra* Benth. *Proceedings of the Indian Academy of Sciences, Section B* 36: 1–11.

Doyle, J. J. and J. L. Doyle. 1987. A rapid DNA isolation procedure for small quantities of fresh leaf tissue. *Phytochemical Bulletin* 19: 11–15.

Doyle, J. J., J. L. Doyle, J. A. Ballenger, E. E. Dickson, T. Kajita, and H. Ohashi. 1997. A phylogeny of the chloroplast gene *rbcL* in the Leguminosae: Taxonomic correlations and insights into the evolution of nodulation. *American Journal of Botany* 84: 541–554.

Drummond, A. J., M. A. Suchard, D. Xie, and A. Rambaut. 2012. Bayesian phylogenetics with BEAUti and the BEAST 1.7. *Molecular Biology and Evolution* 29: 1969–1973.

Dunlop, C. R., G. J. Leach, and I. D. Cowie. 1995. *Flora of the Darwin Region*. Darwin: Conservation Commission of the Northern Territory.

Fabre, P. H., J. M. Brokaw, L. Hufford, M. G. Johnson, and J. J. Schenk. 2025. Resolving phylogenetic relationships of a recent and rapidly evolving clade from western North America (*Mentzelia* section *Bartonia*, Loasaceae). *Systematic Botany* 50: 67–82.

Farruggia, F. T., M. Lavin, and M. F. Wojciechowski. 2018. Phylogenetic systematics and biogeography of the pantropical genus *Sesbania* (Leguminosae). *Systematic Botany* 43: 414–429.

Frahm-Leliveld, J. A. 1953. Some chromosome numbers in tropical leguminous plants. *Euphytica* 2: 46–48.

Frost, L. A., A. M. Bedoya, and L. P. Lagomarsino. 2024. Artifactual orthologs and the need for diligent data exploration in complex phylogenomic datasets: A museomic case study from the Andean flora. *Systematic Biology* 73: 308–322.

Gardner, E. M., M. G. Johnson, J. T. Pereira, A. S. A. Puad, D. Arifiani, Sahromi, N. J. Wickett, and N. J. C. Zerega. 2021. Paralogs and off-target sequences improve phylogenetic resolution in a densely sampled study of the breadfruit genus (*Artocarpus*, Moraceae). *Systematic Biology* 70: 558–575.

GBIF. 2024. Global Biodiversity Information Facility. https://doi.org/10.15468/dl.gyuv22 (last accessed 8 February 2024).

Goldblatt, P. 1981. Cytology and the phylogeny of Leguminosae. *Advances in Legume Systematics* 2: 427–463.

Hale, H., E. Gardner, J. Viruel, L. Pokorny, and M. Johnson. 2020. Strategies for reducing per-sample costs in target capture sequencing for phylogenomics and population genomics in plants. *Applications in Plant Sciences* 8: e11337.

Harvey, M.-A., P. D. Erskine, H. H. Harris, G. K. Brown, E. A. H. Pilon-Smits, L. W. Casey, G. Echevarria, and A. van der Ent. 2020. Distribution and chemical form of selenium in *Neptunia amplexicaulis* from Central Queensland, Australia. *Metallomics* 12: 514–527.

Hei, Z., H. Xiang, J. Zhang, K. Liang, X. Ren, Y. Sun, and R. Wu. 2022. Water mimosa (*Neptunia oleracea* Lour.) can fix and transfer nitrogen to rice in their intercropping system. *Journal of the Science of Food and Agriculture* 102: 156–166.

Hei, Z., H. Xiang, J. Zhang, K. Liang, J. Zhong, M. Li, and X. Ren. 2021. Intercropping of rice and water mimosa (*Neptunia oleracea* Lour.): A novel model to control pests and diseases and improve yield and grain quality while reducing N fertilizer application. *Agriculture* 12: 13.

Hendriks, K. P., T. Mandáková, N. M. Hay, E. Ly, A. H. van Huysduynen, R. Tamrakar, S. K. Thomas, O. Toro-Núñez, J. C. Pires, L. A. Nikolov, M. A. Koch, M. D. Windham, M. A. Lysak, F. Forest, K. Mummenhoff, W. J. Baker, F. Lens, and C. D. Bailey. 2021. The best of both worlds: Combining lineage-specific and universal bait sets in target-enrichment hybridization reactions. *Applications in Plant Sciences* 9: e11438.

Herrera, F., M. R. Carvalho, S. L. Wing, C. Jaramillo, and P. S. Herendeen. 2019. Middle to late Paleocene Leguminosae fruits and leaves from Colombia. *Australian Systematic Botany* 32: 385–408.

Hughes, C. E. and M. Luckow. 2024. *Dichrostachys* Clade. In Bruneau A., L. P. Queiroz, J. J. Ringelberg (Eds.). Advances in Legume Systematics 14. Classification of Caesalpinioideae. Part 2: Higher-level classification. *PhytoKeys* 240: 269–298.

Hughes, C. E., C. D. Bailey, S. Krosnick, and M. A. Luckow. 2003. Relationships among genera of the informal *Dichrostachys* and *Leucaena* groups (Mimosoideae) inferred from nuclear ribosomal ITS sequences. *Advances in Legume Systematics* 10: 221–238.

iDigBio. 2024. Integrated Digitized Biocollections. https://www.idigbio.org/portal (last accessed 19 February 2024).

IPNI. 2025. International Plant Names Index. http://www.ipni.org/ (last accessed June 2025).

James, E. K., M. d. F. Loureiro, A. Pott, V. J. Pott, C. M. Martins, A. A. Franco, and J. I. Sprent. 2001. Flooding-tolerant legume symbioses from the Brazilian Pantanal. *New Phytologist* 150: 723–738.

Johnson, M. G., E. M. Gardner, Y. Liu, R. Medina, B. Goffinet, A. J. Shaw, N. J. C. Zerega, and N. J. Wickett. 2016. HybPiper: Extracting coding sequence and introns for phylogenetics from high-throughput sequencing reads using target enrichment. *Applications in Plant Sciences* 4: 1600016. apps.

Johnson, M. G., L. Pokorny, S. Dodsworth, L. R. Botigué, R. S. Cowan, A. Devault, W. L. Eiserhardt, N. Epitawalage, F. Forest, J. T. Kim, J. H. Leebens-Mack, I. J. Leitch, O. Maurin, D. E. Soltis, P. S. Soltis, G. K. Wong, W. J. Baker, and N. J. Wickett. 2019. A universal probe set for targeted sequencing of 353 nuclear genes from any flowering plant designed using k-medoids clustering. *Systematic Biology* 68: 594–606.

Kadlec, M., D. U. Bellstedt, N. C. L. Maitre, and M. D. Pirie. 2017. Targeted NGS for species level phylogenomics: "made to measure" or "one size fits all? *PeerJ* 5: e3569.

Käss, E. and M. Wink. 1997. Phylogenetic relationships in the Papilionoideae (family Leguminosae) based on nucleotide sequences of cpDNA (*rbc*L) and ncDNA (ITS 1 and 2). *Molecular Phylogenetics and Evolution* 8: 65–88.

Katoh, K. and D. M. Standley. 2013. MAFFT multiple sequence alignment software version 7: Improvements in performance and usability. *Molecular Biology and Evolution* 30: 772–780.

Koenen, E. J. M., C. Kidner, É. R. Souza, M. F. Simon, J. R. Iganci, J. A. Nicholls, G. K. Brown, L. P. de Queiroz, M. Luckow, G. P. Lewis, T. R. Pennington, and C. E. Hughes. 2020a. Hybrid capture of 964 nuclear genes resolves evolutionary relationships in the mimosoid

legumes and reveals the polytomous origins of a large pantropical radiation. *American Journal of Botany* 107: 1710–1735.

Koenen, E. J. M., D. I. Ojeda, F. T. Bakker, J. J. Wieringa, C. Kidner, O. J. Hardy, R. T. Pennington, P. S. Herendeen, A. Bruneau, and C. E. Hughes. 2020b. The origin of the legumes is a complex paleopolyploid phylogenomic tangle closely associated with the Cretaceous–Paleogene (K–Pg) Mass Extinction Event. *Systematic Biology* 70: 508–526.

Koenen, E. J. M., D. I. Ojeda, R. Steeves, J. Migliore, F. T. Bakker, J. J. Wieringa, C. Kidner, O. J. Hardy, R. T. Pennington, A. Bruneau, and C. E. Hughes. 2020c. Large-scale genomic sequence data resolve the deepest divergences in the legume phylogeny and support a near-simultaneous evolutionary origin of all six subfamilies. *New Phytologist* 225: 1355–1369.

Krings, A., W. Wall, and A. Broadhead. 2023. *Parkinsonia*. Pp. 68–71 in *Flora of North America North of Mexico*, vol. 11, Part 1, eds. Flora of North America Editorial Committee. New York and Oxford: Flora of North America Editorial Committee.

Kumar, V. and B. Subramaniam. 1986. *Chromosome Atlas of Flowering Plants of the Indian Subcontinent Volume I: Dicotyledons.* Calcutta: Botanical Survey of India.

Larridon, I., T. Villaverde, A. R. Zuntini, L. Pokorny, G. E. Brewer, N. Epitawalage, I. Fairlie, I. Fairlie, M. Hahn, J. Kim, E. Maguilla, O. Maurin, M. Xanthos, A. L. Hipp, F. Forest, and W. J. Baker. 2020. Tackling rapid radiations with targeted sequencing. *Frontiers in Plant Science* 10: 1655.

Lavin, M., B. P. Schrire, G. Lewis, R. T. Pennington, A. Delgado-Salinas, M. Thulin, C. E. Hughes, A. B. Matos, and M. F. Wojciechowski. 2004. Metacommunity process rather than continental tectonic history better explains geographically structured phylogenies in legumes. *Philosophical Transactions of the Royal Society B: Biological Sciences* 359: 1509–1522.

Lavin, M., P. S. Herendeen, and M. F. Wojciechowski. 2005. Evolutionary rates analysis of Leguminosae implicates a rapid diversification of lineages during the Tertiary. *Systematic Biology* 54: 575–594.

Lewis, G. P. and T. S. Elias. 1981. Mimoseae. Pp. 155–169 in *Advances in Legume Systematics*, Part 1, eds. R. M. Polhill and P. H. Raven. Richmond: Royal Botanic Gardens, Kew.

Lewis, G. P., B. A. Mackinder, and M. Lock. 2005. *Legumes of the World*. Richmond: Royal Botanic Gardens, Kew.

Li, H. and R. Durbin. 2009. Fast and accurate short read alignment with Burrows-Wheeler transform. *Bioinformatics* 25: 1754–1760.

Loureiro, J. 1790. *Flora Cochinchinensis: Sistens plantas in regno Cochinchina nascentes.* Ulyssipone (Lisbon): Typis, et expensis Academicis.

LPWG. 2025. The Legume Phylogeny Working Group. WCVP: Fabaceae checklist (2025v.6) [Data set]. Royal Botanic Gardens, Kew. https://doi.org/10.5281/zenodo.15853560

LPWG. 2017. The Legume Phylogeny Working Group. A new subfamily classification of the Leguminosae based on a taxonomically comprehensive phylogeny. *Taxon* 66: 44–77.

Luckow, M. 1993. Monograph of *Desmanthus* (Leguminosae-Mimosoideae). *Systematic Botany Monographs* 38: 1–166.

Luckow, M. A. 2023. *Dichrostachys*. Pp. 91–92 in *Flora of North America North of Mexico Volume 11, Part 1*, eds. Flora of North America Editorial Committee. New York and Oxford: Flora of North America Editorial Committee.

Luckow, M. and J. Grimes. 1997. A survey of anther glands in the mimosoid legume tribes Parkieae and Mimoseae. *American Journal of Botany* 84: 285–297.

Luckow, M., J. T. Miller, D. J. Murphy, and T. Livshultz. 2003. A phylogenetic analysis of the Mimosoideae (Leguminosae) based on chloroplast DNA sequence data. *Advances in Legume Systematics* 10: 197–220.

Marazzi, B., A. M. Gonzalez, A. Delgado-Salinas, M. A. Luckow, J. J. Ringelberg, and C. E. Hughes. 2019. Extrafloral nectaries in Leguminosae: Phylogenetic distribution, morphological diversity and evolution. *Australian Systematic Botany* 32: 409–458.

Matzke, N. J. 2014. Model selection in historical biogeography reveals that founder-event speciation is a crucial process in island clades. *Systematic Biology* 63: 951–970.

Metcalfe, C. R. 1931. The "aerenchyma" of Sesbania and Neptunia. Bulletin of Miscellaneous Information (Royal Gardens, Kew) 1931: 151–154.

Miège, J. 1960. Troisième liste de nombres chromosomiques d'espèces d'Afrique occidentale. *Annales de la Faculté Des Sciences de L'Université de Dakar* 5: 75–85.

Minh, B. Q., H. A. Schmidt, O. Chernomor, D. Schrempf, M. D. Woodhams, A. von Haeseler, and R. Lanfear. 2020. IQ-TREE 2: New models and efficient methods for phylogenetic inference in the genomic era. *Molecular Biology and Evolution* 37: 1530–1534.

Morales-Briones, D. F., B. Gehrke, C.-H. Huang, A. Liston, H. Ma, H. E. Marx, D. C. Tank, and Y. Yang. 2021. Analysis of paralogs in target enrichment data pinpoints multiple ancient polyploidy events in *Alchemilla* s.l. (Rosaceae). *Systematic Biology* 71: 190–207.

Naturalis. 2024. Naturalis Biodiversity Center Collections. https://www.naturalis.nl (last accessed 19 February 2024).

Nauheimer, L., N. Weigner, E. Joyce, D. Crayn, C. Clarke, and K. Nargar. 2021. HybPhaser: A workflow for the detection and phasing of hybrids in target capture data sets. *Applications in Plant Sciences* 9: e11441.

One Thousand Plant Transcriptomes Initiative. 2019. One thousand plant transcriptomes and the phylogenomics of green plants. *Nature* 574: 679–685.

Pandit, M. K., H. T. W. Tan, and M. S. Bisht. 2006. Polyploidy in invasive plant species of Singapore. *Botanical Journal of the Linnean Society* 151: 395–403.

Paradis, E. and K. Schliep. 2019. Ape 5.0: An environment for modern phylogenetics and evolutionary analyses in R. *Bioinformatics* 35: 526–528.

Peck, J. and B. Serviss. 2011. *Neptunia oleracea* (Fabaceae) new to the continental United States, with new and noteworthy records of several angiosperms in Arkansas. *Journal of the Botanical Research Institute of Texas* 5: 321–326.

Pennell, M., J. M. Eastman, G. J. Slater, J. W. Brown, J. C. Uyeda, R. G. FitzJohn, M. E. Alfaro, and L. J. Harmon. 2014. Geiger v2.0: An expanded suite of methods for fitting macroevolutionary models to phylogenetic trees. *Bioinformatics* 30: 2216–2218.

POWO. 2025. Plants of the World Online. http://www.powo.science.kew.org/ (last accessed June 2025).

R Development Core Team. 2023. *R: A language and environment for statistical computing.* Vienna, Austria: R Foundation for Statistical Computing.

Raman, V. S. and P. C. Kesavan. 1963. Chromosome numbers of a few horticultural plants. *Science and Culture* 29: 609–610.

Rambaut, A., A. J. Drummond, D. Xie, G. Baele, and M. A. Suchard. 2018. Posterior summarization in Bayesian phylogenetics using Tracer 1.7. *Systematic Biology* 67: 901–904.

Ree, R. H. and S. A. Smith. 2008. Maximum likelihood inference of geographic range evolution by dispersal, local extinction, and cladogenesis. *Systematic Biology* 57: 4–14.

Revell, L. J. 2012. phytools: An R package for phylogenetic comparative biology (and other things). *Methods in Ecology and Evolution* 3: 217–223.

Revell, L. J. and L. J. Harmon. 2022. *Phylogenetic Comparative Methods in R.* Princeton, New Jersey: Princeton University Press.

Richardson, A. and K. King. 2008. *Neptunia plena* (Fabaceae: Mimosoideae) rediscovered in Texas. *Journal of the Botanical Research Institute of Texas* 2: 1491–1493.

Ringelberg, J. J., E. J. M. Koenen, J. R. Iganci, L. P. de Queiroz, D. J. Murphy, M. Gaudeul, A. Bruneau, M. Luckow, G. P. Lewis, and C. E. Hughes. 2022. Phylogenomic analysis of 997 nuclear genes reveals the need for extensive generic re-delimitation in Caesalpinioideae (Leguminosae). *PhytoKeys* 205: 3–58.

Ringelberg, J. J., E. J. M. Koenen, B. Sauter, A. Aebli, J. G. Rando, J. R. Iganci, L. P. de Queiroz, D. J. Murphy, M. Gaudeul, A. Bruneau, M. Luckow, G. P. Lewis, J. T. Miller, M. F. Simon, L. S. B. Jordão, M. Morales, C. D. Bailey, M. Nageswara-Rao, J. A. Nicholls, O. Loiseau, R. T. Pennington, K. G. Dexter, N. E. Zimmerman, and C. E. Hughes. 2023. Precipitation is the main axis of tropical plant phylogenetic turnover across space and time. *Science Advances* 9: eade4954.

Rowe, K. C., A. S. Achmadi, P. Fabre, J. J. Schenk, S. J. Steppan, and J. A. Esselstyn. 2019. Oceanic islands of Wallacea as a source for dispersal and diversification of murine rodents. *Journal of Biogeography* 46: 2752–2768.

Sarkar, A. K., N. Datta, R. Mallick, and U. Chatterjee. 1976. IOPB chromosome number reports LIV. *Taxon* 25: 631–649.

Saunders, T. C., I. Larridon, W. J. Baker, R. L. Barrett, F. Forest, E. Françoso, O. Maurin, S. Rokni, and E. H. Roalson. 2024. Tangled webs and spider-flowers: Phylogenomics, biogeography, and seed morphology inform the evolutionary history of Cleomaceae. *American Journal of Botany* 111: e16399.

Schenk, J. J. 2016. Consequences of secondary calibrations on divergence time estimates. *Plos One* 11: e0148228.

Schenk, J. J. and L. Hufford. 2010. Effects of substitution models on divergence time estimates: Simulations and an empirical study of model uncertainty using Cornales. *Systematic Botany* 35: 578–592.

Schenk, J. J., L. E. Becklund, S. J. Carey, and P. P. Fabre. 2023. What is the "modified" CTAB protocol? Characterizing modifications to the

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CTAB DNA extraction protocol. *Applications in Plant Sciences* 11: e11517.

Seago, J. L., L. C. Marsh, K. J. Stevens, A. Soukup, O. Votrubová, and D. E. Enstone. 2005. A re-examination of the root cortex in wetland flowering plants with respect to aerenchyma. *Annals of Botany* 96: 565–579.

SEINet. 2024. SEINet Portal Network. https://swbiodiversity.org/seinet (last accessed 17 February 2024).

Shah, T., J. V. Schneider, G. Zizka, O. Maurin, W. Baker, F. Forest, G. E. Brewer, V. Savolainen, I. Darbyshire, and I. Larridon. 2021. Joining forces in Ochnaceae phylogenomics: A tale of two targeted sequencing probe kits. *American Journal of Botany* 108: 1201–1216.

Shavvon, R. S., K. S. Osaloo, A. A. Maassoumii, F. Moharrek, S. K. Erkul, A. R. Lemmon, E. M. Lemmon, I. Michalak, A. N. Muellner-Riehl, and A. Favre. 2017. Increasing phylogenetic support for explosively radiating taxa: The promise of high-throughput sequencing for *Oxytropis* (Fabaceae). *Journal of Systematics and Evolution* 55: 385–404.

Shen, W., S. Le, Y. Li, and F. Hu. 2016. SeqKit: A cross-platform and ultra-fast toolkit for FASTA/Q file manipulation. *Plos One* 11: e0163962.

Shimodaira, H. 2002. An approximately unbiased test of phylogenetic tree selection. *Systematic Biology* 51: 492–508.

Silva, J. S., M. S. Carvalho, G. S. Santos, F. T. Braga, M. J. Gomes de Andrade, and V. de Freitas Mansano. 2020. *Neptunia windleriana*: A new polyploid species of *Neptunia* (Leguminosae) from Brazil recognized by anatomy, morphology and cytogenetics. *Systematic Botany* 45: 483–494.

Simon, M. F., R. Grether, L. P. de Queiroz, T. E. Särkinen, V. F. Dutra, and C. E. Hughes. 2011. The evolutionary history of *Mimosa* (Leguminosae): Toward a phylogeny of the sensitive plants. *American Journal of Botany* 98: 1201–1221.

Siniscalchi, C. M., O. Hidalgo, L. Palazzesi, J. Pellicer, L. Pokorny, O. Maurin, I. J. Leitch, F. Forest, W. J. Baker, and J. R. Mandel. 2021. Lineage-specific vs. universal: A comparison of the Compositae1061 and Angiosperms353 enrichment panels in the sunflower family. *Applications in Plant Sciences* 9: e11422.

Sirichamorn, Y., D. C. Thomas, F. A. C. B. Adema, and P. C. van Welzen. 2014. Historical biogeography of *Aganope*, *Brachypterum* and *Derris* (Fabaceae, tribe Millettieae): Insights into the origins of Palaeotropical intercontinental disjunctions and general biogeographical patterns in Southeast Asia. *Journal of Biogeography* 41: 882–893.

Slater, G. S. C. and E. Birney. 2005. Automated generation of heuristics for biological sequence comparison. *BMC Bioinformatics* 6: 31.

Smith, S. A., J. W. Brown, and J. F. Walker. 2018. So many genes, so little time: A practical approach to divergence-time estimation in the genomic era. *PLoS ONE* 13: e0197433.

Stamatakis, A. 2014. RAxML Version 8: A tool for phylogenetic analysis and post-analysis of large phylogenies. *Bioinformatics* 30: 1312–1313.

Sulaiman, S. F., A. Culham, and J. B. Harborne. 2003. Molecular phylogeny of Fabaceae based on *rbcL* sequence data: With special emphasis on the tribe Mimoseae (Mimosoideae). *Asia Pacific Journal of Molecular Biology and Biotechnology* 11: 9–35.

Tabosa, A. B., L. Q. Matias, and F. R. Martins. 2012. Live fast and die young: The aquatic macrophyte dynamics in a temporary pool in the Brazilian semiarid region. *Aquatic Botany* 102: 71–78.

Thiers, B. M. 2025 (continuously updated). Index Herbariorum. The William & Lynda Steere Herbarium. https://sweetgum.nybg.org/science/ih/ (last accessed May 2025).

Thomas, A. E., J. Igea, H. M. Meudt, D. C. Albach, W. G. Lee, and A. J. Tanentzap. 2021. Using target sequence capture to improve the phylogenetic resolution of a rapid radiation in New Zealand *Veronica*. *American Journal of Botany* 108: 1289–1306.

Thulin, M. 2023. On *Calliandra* and *Afrocalliandra* (Fabaceae-Caesalpinioideae). *Phytotaxa* 595: 1–6.

Tseng, Y.-H., L.-Y. Kuo, I. Borokini, and S. Fawcett. 2024. The role of deep hybridization in fern speciation: Examples from the Thelypteridaceae. *American Journal of Botany* 111: e16388.

Tucker, S. C. 1988. Heteromorphic flower development in *Neptunia pubescens*, a mimosoid legume. *American Journal of Botany* 75: 205–224.

Turner, B. L. 1951. Revision of the United States species of *Neptunia* (Leguminosae). *American Midland Naturalist* 46: 82–92.

Turner, B. L. and O. S. Fearing. 1960. The basic chromosome number of the genus *Neptunia* (Leguminosae-Mimosoideae). *Madroño* 15: 184–187.

Tyndall, R. W. 2011. Long-term monitoring of two subpopulations of the federally threatened *Aeschynomene virginica* (Sensitive Joint-vetch) in Maryland. *Castanea* 76: 20–27.

Ufimov, R., V. Zeisek, S. Píšová, W. J. Baker, T. Fér, M. van Loo, C. Dobeš, and R. Schmickl. 2021. Relative performance of customized and universal probe sets in target enrichment: A case study in subtribe Malinae. *Applications in Plant Sciences* 9: e11442.

Van der Ent, A., M. Salinitro, D. Brueckner, K. M. Spiers, S. Montanari, A. Tassoni, and M. Schiavon. 2023. Differences and similarities in selenium biopathways in *Astragalus*, *Neptunia* (Fabaceae) and *Stanleya* (Brassicaceae) hyperaccumulators. *Annals of Botany* 132: 349–361.

Vatanparast, M., A. Powell, J. J. Doyle, and A. N. Egan. 2018. Targeting legume loci: A comparison of three methods for target enrichment bait design in Leguminosae phylogenomics. *Applications in Plant Sciences* 6: e1036.

Warren, D. L., A. J. Geneva, and R. Lanfear. 2017. RWTY (R We There Yet): An R package for examining convergence of Bayesian phylogenetic analyses. *Molecular Biology and Evolution* 34: 1016–1020.

Wheeler, J. R. 1992. Mimosaceae. In *Flora of the Kimberley Region*, ed. J. R. Wheeler. Como, Western Australia: Department of Conservation and Land Management, Como.

Whibley, D. J. 1981. *Neptunia*. Pp. 114 in *Flora of Central Australia*, ed. J. P. Jessop. Sydney: Reed Books Pty Ltd.

Windler, D. R. 1966. A revision of the genus *Neptunia* (Leguminosae). *Australian Journal of Botany* 14: 379–420.

Wing, S. L., F. Herrera, C. A. Jaramillo, C. Gómez-Navarro, P. Wilf, and C. C. Labandeira. 2009. Late Paleocene fossils from the Cerrejón Formation, Colombia, are the earliest record of neotropical rainforest. *Proceedings of the National Academy of Sciences* 106: 18627–18632.

Wojciechowski, M. F., M. Lavin, and M. J. Sanderson. 2004. A phylogeny of legumes (Leguminosae) based on analysis of the plastid *mat K* gene resolves many well-supported subclades within the family. *American Journal of Botany* 91: 1846–1862.

Yang, Y. and S. A. Smith. 2014. Orthology inference in nonmodel organisms using transcriptomes and low-coverage genomes: Improving accuracy and matrix occupancy for phylogenomics. *Molecular Biology and Evolution* 31: 3081–3092.

Zhang, C., M. Rabiee, E. Sayyari, and S. Mirarab. 2018. ASTRAL-III: Polynomial time species tree reconstruction from partially resolved gene trees. *BMC Bioinformatics* 19: 15–30.

Appendix 1. Voucher specimens of tissues included in the study. Order of the data are the species, collector name and number, herbarium (following Index Herbariorum; Thiers 2025) or "GenBank" for accessions downloaded from GenBank, and NCBI Short Read Archive BioSample number.

Ingroup: *Neptunia amplexicaulis* Domin, *A.R. Bean 13356*, BRI, Australia, SAMN50644672; *Neptunia amplexicaulis* f. *richmondii* Windler, *Byrnes 3044*, BRI, Australia, SAMN50644673; *Neptunia amplexicaulis* Domin, *Fensham 5856*, BRI, Australia, SAMN50644674; *Neptunia dimorphantha* Domin, *B.M. Waterhouse 5175*, CANB, Australia, SAMN50644676; *Neptunia dimorphantha* Domin, *D.L. Lewis 1912*, NSW (GenBank), Australia, SAMEA11459815; *Neptunia dimorphantha* Domin, *D.W. Rust 63K*, CANB, Australia, SAMN50644677; *Neptunia dimorphantha* Domin (*Neptunia gracilis* f. *glandulosa* Windler), *B. Wirf 722*, DNA, Australia, SAMN50644675; *Neptunia dimorphantha* Domin (*Neptunia gracilis* f. *glandulosa* Windler), *G.M. Wightman 4977*, DNA, Australia, SAMN50644678; *Neptunia dimorphantha* Domin (*Neptunia gracilis* f. *glandulosa* Windler), *M. Evans 3201*, DNA, Australia, SAMN50644679; *Neptunia gracilis* Benth., *A.R. Bean 2633*, BRI, Australia, SAMN50644680; *Neptunia gracilis* Benth., *D. Halford Q7689A*, BRI, Australia, SAMN50644681; *Neptunia gracilis* Benth., *L. Johnson D 722*, Z, Australia, SAMN50644682; *Neptunia gracilis* Benth., *M. Larazarides 4160*, US, Australia, SAMN50644683; *Neptunia heliophila* A.R.Bean, *I.R. Telford 11469*, CBG, Australia, SAMN50644684; *Neptunia heliophila* A.R.Bean, *J.R. Hosking 2177*, CANB, Australia, SAMN50644685; *Neptunia heliophila* A.R.Bean, *J.R. Manocochie 2305*, TEX, Australia, SAMN50644686; *Neptunia hispida* A.R.Bean, *G. Leach 2941*, BRI, Australia, SAMN50644687; *Neptunia hispida* A.R.Bean, *M. Lazarides 7779*, CANB, Australia, SAMN50644688; *Neptunia insignis* A.R.Bean, *A.R. Bean 18740*, BRI, Australia, SAMN50644689; *Neptunia insignis* A.R.Bean, *I.G. Champion 2139*, BRI, Australia, SAMN50644690; *Neptunia javanica* Miq., *O.E. White 20*, US, Myanmar, SAMN50644691; *Neptunia javanica* Miq., *P.E. Schmutz 34*, L, Indonesia, SAMN50644692; *Neptunia javanica* Miq., *R. Pooma 5491*, L, Thailand, SAMN50644693; *Neptunia longipila* A.R.Bean, *S. van Leeuwen 5099*, CANB, Australia, SAMN50644694; *Neptunia lutea* (Leavenw.) Benth., *D.J. Rosen 4801*, TEX, United States, SAMN50644696; *Neptunia lutea* (Leavenw.) Benth., *B. Lipscomb 2381*, TEX, United States,

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SAMN50644695; *Neptunia lutea* (Leavenw.) Benth., *F.R. Waller 2706*, US, United States, SAMN50644697; *Neptunia lutea* (Leavenw.) Benth., *L.E. Becklund 62*, BHO, United States, SAMN50644698; *Neptunia lutea* (Leavenw.) Benth., *R. Kral 39772*, TEX, Alabama, SAMN50644699; *Neptunia major* Windler, *A.R. Bean 24919*, CANB, Australia, SAMN50644700; *Neptunia major* Windler, *B.M. Stuckey 414*, MO, Australia, SAMN50644701; *Neptunia major* Windler, *H. Demarz 8836*, CANB, Australia, SAMN50644702; *Neptunia monosperma* F.Muell. ex Benth., *A. Fraser 363*, CANB, Australia, SAMN50644703; *Neptunia monosperma* F.Muell. ex Benth., *D.L. Lewis 1068*, MO, Australia, SAMN50644704; *Neptunia monosperma* F.Muell. ex Benth., *D.L. Lewis 2008*, CANB, Australia, SAMN50644705; *Neptunia monosperma* F.Muell. ex Benth., *L.E. Becklund 68*, BHO, Australia, SAMN50644706; *Neptunia oleracea* Lour., *B. Rothschild 95*, TEX, Mexico, SAMN50644707; *Neptunia oleracea* Lour., *C. Rakotovao 6650*, MO, Madagascar, SAMN50644708; *Neptunia oleracea* Lour., *J.P. Essou 1580*, MO, Benin, SAMN50644709; *Neptunia oleracea* Lour., *J.W. Nowicke 413*, MO, Sri Lanka, SAMN50644710; *Neptunia oleracea* Lour., *K. Swatabaudbe 36*, MO, Thailand, SAMN50644711; *Neptunia oleracea* Lour., *L. Acosta 2519*, MO, Costa Rica, SAMN50644712; *Neptunia oleracea* Lour., *L. Festo 2315*, MO, Kenya, SAMN50644713; *Neptunia oleracea* Lour., *M. Rimachi Y. 10494*, TEX, Peru, SAMN50644714; *Neptunia oleracea* Lour., *P.A. Smith 3810*, MO, Botswana, SAMN50644715; *Neptunia oleracea* Lour., *R.K. Grosvenor 592*, MO, Zimbabwe, SAMN50644716; *Neptunia paucijuga* A.R.Bean, *C.R. Michell 3954*, DNA, Australia, SAMN50644717; *Neptunia paucijuga* A.R.Bean, *N.J. Cuff 662*, DNA, Australia, SAMN50644718; *Neptunia plena* (L.) Benth., *A.R. Novelo 4253*, Z, Mexico, SAMN50644719; *Neptunia plena* (L.) Benth., *D. Burch 6673*, MO, Puerto Rico, SAMN50644720; *Neptunia plena* (L.) Benth., *G.E. Crow 9600*, MO, Costa Rica, SAMN50644722; *Neptunia plena* (L.) Benth., *G. Hatschbach 60910*, Z, Brazil, SAMN50644721; *Neptunia plena* (L.) Benth., *G.R. Proctor 27727*, TEX, Jamaica, SAMN50644723; *Neptunia plena* (L.) Benth., *I.L. Wiggins 512*, US, Ecuador, SAMN50644724; *Neptunia plena* (L.) Benth., *J. Vitelli WM198*, BRI, Australia, SAMN50644725; *Neptunia plena* (L.) Benth., *J. Westaway 3786*, CANB, Australia, SAMN50644726; *Neptunia plena* (L.) Benth., *L.E. Becklund 52*, BHO, United States, SAMN50644728; *Neptunia plena* (L.) Benth., L. *O'Bryan s.n.*, BRI, Australia, SAMN50644727; *Neptunia plena* (L.) Benth., *M.J. Jansen-Jacobs 3980*, MO, Guyana, SAMN50644729; *Neptunia plena* (L.) Benth., *P.S. Bentley 277*, US, Ecuador, SAMN50644730; *Neptunia plena* (L.) Benth., *R.D. Worthington 12337*, MO, Malaysia, SAMN50644732; *Neptunia plena* (L.) Benth., *R.E. Jackson V-35*, TEX, Venezuela, SAMN50644733; *Neptunia plena* (L.) Benth., *R. Gonto 06173*, MO, Venezuela, SAMN50644731; *Neptunia plena* (L.) Benth., *T.R. van Devender 92-1100*, TEX, Mexico, SAMN50644735; Neptunia proxima A.R.Bean, *K.R. McDonald 17070*, BRI, Australia, SAMN50644736; *Neptunia proxima* A.R.Bean, *K.R. McDonald 18188*, BRI, Australia, SAMN50644737; *Neptunia pubescens* Benth., *A.N. Egan 4276*, US, United States, SAMN50644738; *Neptunia pubescens* Benth., *E.P. Killip 41174*, US, United States, SAMN50644739; *Neptunia pubescens* Benth., *Gillis 8073*, A, United States, SAMN50644741; *Neptunia pubescens* Benth., *G. Seijo 2411*, TEX, Uruguay, SAMN50644740; *Neptunia pubescens* Benth., *J.d.J. Jimenez 4081*, US, Dominican Republic, SAMN50644743; *Neptunia pubescens* Benth., *J. Hudson 941*, MO, Peru, SAMN50644742; *Neptunia pubescens* Benth., *J.J. Schenk 2627*, BHO, United States, SAMN50644744; *Neptunia pubescens* Benth., *J.J. Schenk 2678*, BHO, United States, SAMN50644745; *Neptunia pubescens* Benth., *L.E. Becklund 46*, BHO, United States, SAMN50644746; *Neptunia pubescens* Benth., *L.E. Becklund 53*, BHO, United States, SAMN50644747; *Neptunia pubescens* Benth., *L.E. Becklund 54*, BHO, United States, SAMN50644748; *Neptunia pubescens* Benth., *L.E. Becklund 65*, BHO, United States, SAMN50644749; *Neptunia pubescens* Benth., *L.E. Becklund 66*, BHO, United States, SAMN50644750; *Neptunia pubescens* Benth., *O. Ahumada 1743*, MO, Argentina, SAMN50644751; *Neptunia pubescens* var. *microcarpa* (Rose) Windler, *L.E. Becklund 57*, BHO, United States, SAMN50644753; *Neptunia pubescens* var. *microcarpa* (Rose) Windler, *J. Rzedowski 38840*, TEX, Mexico, SAMN50644752; *Neptunia scutata* A.R.Bean, *D.L. Lewis 2661*, MO, Australia, SAMN50644754; *Neptunia scutata* A.R.Bean, *J. Egan 2995*, CANB, Australia, SAMN50644755; *Neptunia scutata* A.R.Bean, *P.R. Foulkes 421*, CANB, Australia, SAMN50644756; *Neptunia scutata* A.R.Bean, *R.W. Purdie 4866*, CANB, Australia, SAMN50644757; *Neptunia tactilis* A.R.Bean, *J.R. Clarkson 6043*, CANB, Australia, SAMN50644758; *Neptunia tactilis* A.R.Bean, *J.R. Manocochie 2168*, CANB, Australia, SAMN50644759; *Neptunia triquetra* Benth., *L.G.J. van der Maesen 3867*, L, India, SAMN50644760; *Neptunia valida* A.R.Bean, *A. Roberts APR1079*, CANB, Australia, SAMN50644761; *Neptunia valida* A.R.Bean, *R.W. Purdie 3309*, CBG, Australia, SAMN50644762; *Neptunia windleriana* Santos-Silva & Mansano, *R.M. Harley 19075*, US, Brazil, SAMN50644734; *Neptunia xanthonema* A.R.Bean, *A.V. Slee 3332*, CANB, Australia, SAMN50644763; *Neptunia xanthonema* A.R.Bean, *R.W. Purdie 2931*, CBG, Australia, SAMN50644764.

Outgroup: *Acaciella angustissima* (Mill.) Britton & Rose, *L.E. Becklund 61*, BHO, United States, SAMN50644664; *Albizia julibrissin* Durazz., *L.E. Becklund 67*, BHO, United States, SAMN50644665; *Calliandra eriophylla* Benth., *L.E. Becklund 73*, BHO, United States, SAMN50644666; *Desmanthus velutinus* Scheele, *L.E. Becklund 58*, BHO, United States, SAMN50644667; *Desmanthus virgatus* (L.) Willd., *L.E. Becklund 48*, BHO, United States, SAMN50644668; *Dichrostachys cinerea* (L.) Wight & Arn., *M. Botha 14*, K (GenBank), Unknown, SAMEA11052210; *Lemurodendron capuronii* Villiers & P.Guinet, *E.J.M. Koenen 435*, Z, Madagascar, SAMN50644669; *Mimosa aculeaticarpa* Ortega, *L.E. Becklund 55*, BHO, United States, SAMN50644670; *Neltuma glandulosa* (Torr.) Britton & Rose, *L.E. Becklund 49*, BHO, United States, SAMN50644671; *Parkinsonia aculeata* L., *L.E. Becklund 56*, BHO, United States, SAMN50644765; *Prosopidastrum mexicanum* (Dressler) Burkart, *R. Moran 17061*, TEX, Mexico, SAMN50644766.

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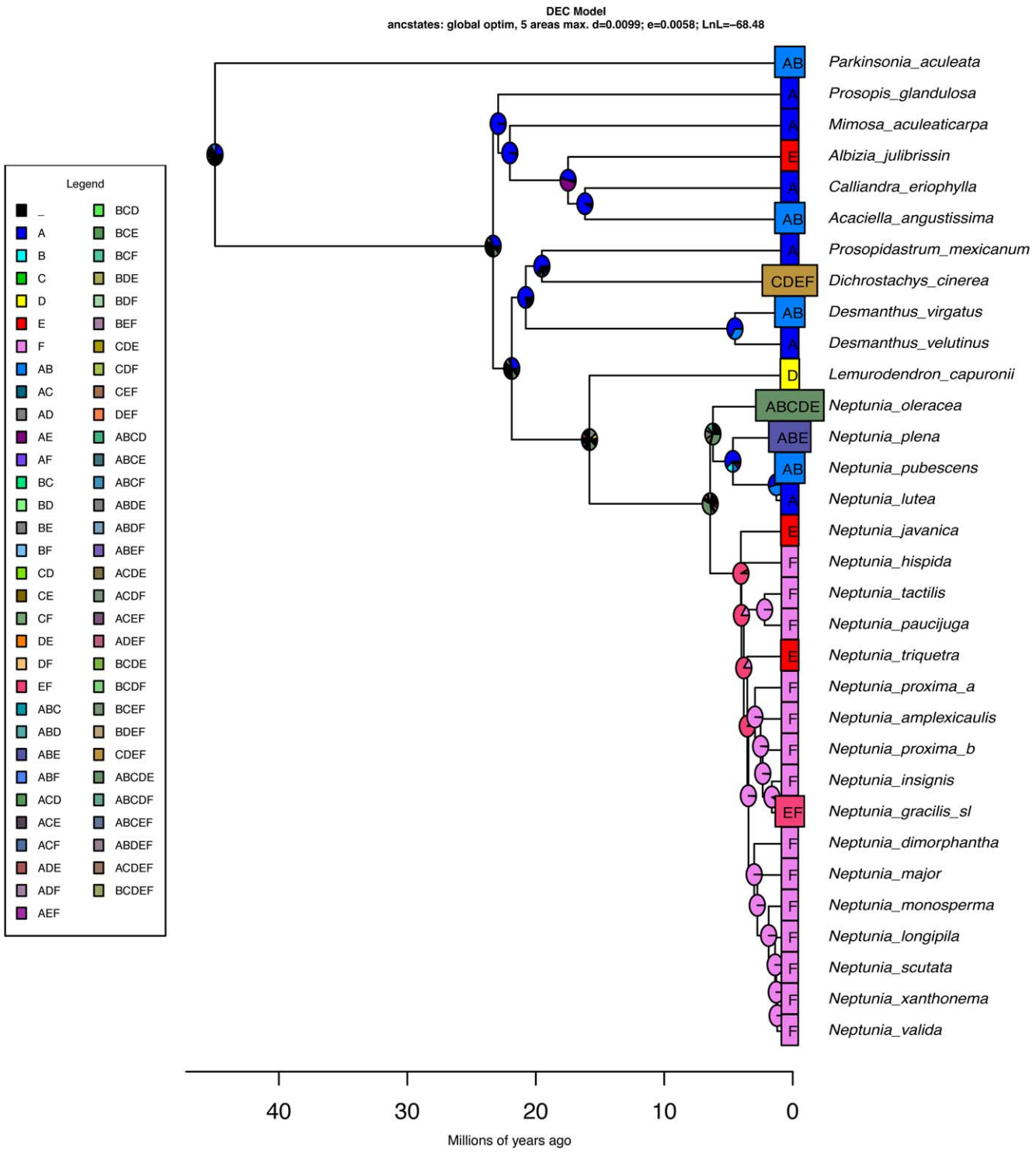

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FIG. S1. The historical biogeographic estimations of *Neptunia* based on the DEC model, with the widespread *N. oleracea* and *N. plena* assigned to their full geographic ranges.

TABLE S1. Accepted *Neptunia* species with heterotypic synonyms. Current taxa include those described in Windler (1966), Silva et al. (2020), and Bean (2022). Synonyms were determined from previous descriptions, the International Plant Names Index (IPNI 2025), and Plants of the World Online (POWO 2025). Heterotypic synonyms that were not included in previous revisions and that were incorrectly attributed, dubious, or unverifiable from original descriptions or digitized type specimens were not included here. The list includes only the oldest heterotypic synonyms.

Accepted species	Synonymous taxa
Neptunia dimorphantha	*Neptunia gracilis* f. *glandulosa* Windler
Neptunia gracilis	*Neptunia depauperata* Merr.
Neptunia lutea	*Neptunia tenuis* Benth.
	Neptunia lutea var. *multipinnata* B.L.Turner
Neptunia oleracea	*Desmanthus stolonifer* DC.
	Mimosa aquatica Pers.
	Mimosa lacustris Bonpl.
	Mimosa prostrata Lam.
	Neptunia natans (L.f.) Druce
Neptunia plena	*Desmanthus comosus* A.Rich.
	Desmanthus polyphyllus DC.
	Mimosa adenanthera Roxb.
	Mimosa lycopodioides Desv.
	Mimosa punctata L.
	Neptunia plena f. *lumbricoides* Fassett
	Neptunia polyphylla Benth.
	Neptunia surinamensis Steud.
Neptunia pubescens var. *pubescens*	*Neptunia floridana* Small
	Neptunia lindheimeri B.L.Rob.
	Neptunia mazatlana S.Watson
	Neptunia palmeri S.Watson
Neptunia pubescens var. *microcarpa* (Rose) Windler	*Neptunia microcarpa* Rose
Neptunia triquetra	*Mimosa natans* L.f.
Neptunia heliophila	*Neptunia gracilis* var. *villosula* Benth.
Neptunia hispida	*Neptunia* 'Darwin,' *pro parte* (Dunlop et al. 1995)
Neptunia javanica	*Desmanthus pulsillus* Zipp. ex Span.
	Desmanthus acinaciformis Span.
	Neptunia robertsonii Raizada
Neptunia scutata	*Neptunia dimorphantha* auct. non Domin, *pro parte* (Whibley 1981)
Neptunia tactilis	*Neptunia* 'Darwin,' *pro parte* (Dunlop et al. 1995)
Neptunia valida	*Neptunia dimorphantha* auct. non Domin, *pro parte* (Wheeler 1992)
Neptunia xanthonema	*Neptunia dimorphantha* var. *clementii* Domin
	Neptunia monosperma auct. non Domin (Black 1948)

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TABLE S2. Laboratory work and sequencing read assembly information.

Species	Collector and number	Sequencing run	Input read pairs	Trimmomatic read pairs	Trimmomatic clip length	Hybpiper reads mapped	Genes with length ≥ 50%	Instances removed with PhylteR
Acaciella angustissima (Mill.) Britton & Rose	L. E. Becklund 61	NovaSeq1	12,735,902	11,741,211	57	4,500,413	310	2
Albizia julibrissin Durazz.	L. E. Becklund 67	NovaSeq1	16,374,331	15,327,110	57	5,517,460	325	1
Calliandra eriophylla Benth.	L. E. Becklund 73	NovaSeq2	5,689,189	4,909,822	130	6,203,838	331	1
Desmanthus velutinus Scheele	L. E. Becklund 58	NovaSeq1	16,403,429	15,406,647	57	8,510,301	331	2
Desmanthus virgatus (L.) Willd.	L. E. Becklund 48	NovaSeq1	40,564,217	38,178,249	57	28,529,701	304	1
Dichrostachys cinerea (L.) Wight & Arn.	M. Botha 14	GenBank	1,068,450	1,068,450	57	337,512	292	2
Lemurodendron capuronii Villiers & P.Guinet	E.J. M. Koenen 435	NovaSeq2	3,337,629	2,914,252	130	4,628,710	338	2
Mimosa aculeaticarpa Ortega	L. E. Becklund 55	NovaSeq1	25,433,669	23,853,808	57	14,226,658	334	4
Neltuma glandulosa (Torr.) Britton & Rose	L. E. Becklund 49	NovaSeq1	17,214,264	16,087,429	57	6,651,453	332	1
Neptunia amplexicaulis Domin	A. R. Bean 13356	MiSeq	1,643,478	1,475,207	57	2,233,574	325	0
Neptunia amplexicaulis f. *richmondii* Windler	Byrnes 3043	MiSeq	812,679	741,331	57	971,023	254	1
Neptunia amplexicaulis Domin	Fensham 5856	MiSeq	1,953,259	1,714,559	57	2,671,984	331	1
Neptunia dimorphantha Domin	B. M. Waterhouse 5175	NovaSeq2	2,074,125	1,871,757	57	2,594,037	281	6
Neptunia dimorphantha Domin	D. L. Lewis 1912	GenBank	6,901,018	5,816,261	57	3,231,215	330	1
Neptunia dimorphantha Domin	D. W. Rust 63K	NovaSeq2	8,575,118	7,616,190	57	1,533,199	284	2
Neptunia gracilis Benth.	A. R. Bean 2633	MiSeq	1,004,507	926,422	57	1,359,510	264	5
Neptunia gracilis Benth.	D. Halford Q7689A	MiSeq	3,420,634	2,973,480	57	4,619,344	336	1
Neptunia gracilis Benth.	L. Johnson D 722	NovaSeq2	9,256,479	6,186,595	130	2,108,265	324	1
Neptunia gracilis Benth.	M. Larazarides 4160	NovaSeq2	9,862,428	5,676,094	130	2,090,559	319	1
Neptunia gracilis f. *glandulosa* Windler	B. Wirf 722	NovaSeq2	9,015,022	7,717,791	130	12,006,147	334	0
Neptunia gracilis f. *glandulosa* Windler	G. M. Wightman 4977	NovaSeq2	7,091,569	5,797,682	130	9,343,353	333	0
Neptunia gracilis f. *glandulosa* Windler	M. Evans 3201	NovaSeq2	8,976,715	7,591,729	130	12,451,980	335	1
Neptunia heliophila A.R.Bean	I. R. Telford 11469	NovaSeq2	5,357,728	4,407,222	130	6,994,753	335	1
Neptunia heliophila A.R.Bean	J. R. Hosking 2177	NovaSeq2	6,643,968	5,764,939	130	9,372,352	331	0
Neptunia heliophila A.R.Bean	J. R. Manocochie 2305	NovaSeq1	14,026,714	13,076,054	57	6,611,701	310	7
Neptunia hispida A.R.Bean	G. Leach 2941	MiSeq	1,510,359	1,341,466	57	2,000,720	313	2
Neptunia hispida A.R.Bean	M. Lazarides 7779	NovaSeq2	5,960,902	4,110,388	130	6,622,553	328	1
Neptunia insignis A.R.Bean	A. R. Bean 18740	MiSeq	2,152,643	1,930,523	57	3,049,220	324	0
Neptunia insignis A.R.Bean	I. G. Champion 2139	MiSeq	2,621,847	2,327,075	57	3,756,045	329	0
Neptunia javanica Miq.	O. E. White 20	NovaSeq2	4,053,444	3,708,657	57	4,315,955	287	4
Neptunia javanica Miq.	P. E. Schmutz 34	NovaSeq2	4,743,990	2,692,552	57	300,454	87	0
Neptunia javanica Miq.	R. Pooma 5491	NovaSeq2	2,032,889	1,756,463	57	1,656,217	245	3
Neptunia longipila A.R.Bean	S. van Leeuwen 5099	NovaSeq2	5,920,884	4,882,230	130	7,749,862	339	0
Neptunia lutea (Leavenw.) Benth.	D. J. Rosen 4801	NovaSeq2	47,298,854	37,403,896	130	13,041,912	343	0
Neptunia lutea (Leavenw.) Benth.	Lipscomb 2381	NovaSeq2	2,410,844	2,177,198	57	2,663,464	299	0
Neptunia lutea (Leavenw.) Benth.	F. R. Waller 2706	NovaSeq2	7,682,038	4,946,253	130	1,754,528	330	2
Neptunia lutea (Leavenw.) Benth.	L. E. Becklund 62	NovaSeq1	22,343,737	21,054,623	57	11,000,895	341	0
Neptunia lutea (Leavenw.) Benth.	R. Kral 39772	NovaSeq1	21,252,851	18,293,664	57	6,243,076	240	3
Neptunia major Windler	A. R. Bean 24919	NovaSeq2	762,941	702,978	57	788,349	127	1
Neptunia major Windler	B. M. Stuckey 414	NovaSeq2	8,079,463	5,090,756	130	1,749,391	318	0
Neptunia major Windler	H. Demarz 8836	NovaSeq2	10,816,548	6,234,933	130	2,060,966	314	0
Neptunia monosperma F.Muell. ex Benth.	A. Fraser 363	NovaSeq2	6,760,639	4,452,793	130	6,420,003	290	1
Neptunia monosperma F.Muell. ex Benth.	D. L. Lewis 1068	NovaSeq2	9,980,226	8,568,775	130	12,560,647	337	2
Neptunia monosperma F.Muell. ex Benth.	D. L. Lewis 2008	NovaSeq2	11,362,884	9,952,068	130	16,213,353	330	0
Neptunia monosperma F.Muell. ex Benth.	L. E. Becklund 68	NovaSeq1	11,343,021	10,596,482	57	4,098,083	322	3
Neptunia oleracea Lour.	B. Rothschild 95	NovaSeq1	15,471,089	14,461,849	57	9,823,864	327	4
Neptunia oleracea Lour.	C. Rakotovao 6650	NovaSeq2	9,615,294	8,043,080	57	1,912,308	299	0
Neptunia oleracea Lour.	J. P. Essou 1580	NovaSeq2	3,954,103	3,261,824	57	658,992	257	1

(*Continued*)

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TABLE S2. (CONTINUED).

Species	Collector and number	Sequencing run	Input read pairs	Trimmomatic read pairs	Trimmomatic clip length	Hybpiper reads mapped	Genes with length ≥ 50%	Instances removed with PhylteR
Neptunia oleracea Lour.	J. W. Nowicke 413	NovaSeq2	217,210	120,672	57	141,769	17	0
Neptunia oleracea Lour.	K. Swatabaudbe 36	NovaSeq2	2,323,542	1,053,478	57	185,696	66	0
Neptunia oleracea Lour.	L. Acosta 2519	NovaSeq2	2,897,836	2,008,762	57	1,807,785	80	2
Neptunia oleracea Lour.	L. Festo 2315	NovaSeq2	2,710,911	1,932,090	130	2,536,702	283	0
Neptunia oleracea Lour.	M. Rimachi Y. 10494	NovaSeq1	1,206,219	10,205,788	57	4,552,504	304	2
Neptunia oleracea Lour.	P. A. Smith 3810	NovaSeq2	2,926,751	2,497,694	57	2,232,776	91	4
Neptunia oleracea Lour.	R. K. Grosvenor 592	NovaSeq2	8,558,432	7,861,438	57	8,521,595	241	2
Neptunia paucijuga A.R.Bean	C. R. Michell 3954	NovaSeq2	7,466,166	5,929,695	130	9,052,913	328	2
Neptunia paucijuga A.R.Bean	N. J. Cuff 662	NovaSeq2	9,202,261	7,895,823	130	12,288,076	334	0
Neptunia plena (L.) Benth.	A. R. Novelo 4253	NovaSeq2	7,809,528	6,818,046	130	11,057,316	337	2
Neptunia plena (L.) Benth.	D. Burch 6673	NovaSeq2	5,205,100	4,604,414	57	4,405,825	142	3
Neptunia plena (L.) Benth.	G. E. Crow 9600	NovaSeq2	3,761,941	3,127,983	57	786,846	137	3
Neptunia plena (L.) Benth.	G. Hatschbach 60910	NovaSeq2	5,360,695	3,921,603	57	749,159	130	1
Neptunia plena (L.) Benth.	G. R. Proctor 27727	NovaSeq1	12,980,496	11,956,871	57	3,968,354	273	4
Neptunia plena (L.) Benth.	I. L. Wiggins 512	NovaSeq2	6,076,781	5,190,587	57	918,408	268	3
Neptunia plena (L.) Benth.	J. Vitelli WM198	MiSeq	2,271,314	2,010,458	57	3,275,481	334	0
Neptunia plena (L.) Benth.	J. Westaway 3786	NovaSeq2	5,857,912	4,750,228	130	7,898,316	332	1
Neptunia plena (L.) Benth.	L. E. Becklund 52	NovaSeq1	15,772,603	14,850,213	57	7,790,537	330	2
Neptunia plena (L.) Benth.	L. O'Bryan s.n.	MiSeq	2,189,826	1,925,155	57	3,008,985	333	0
Neptunia plena (L.) Benth.	M. J. Jansen-Jacobs 3980	NovaSeq2	3,200,271	2,832,340	57	3,710,192	275	5
Neptunia plena (L.) Benth.	P. S. Bentley 277	NovaSeq2	6,977,957	5,642,918	130	9,169,886	341	1
Neptunia plena (L.) Benth.	R. D. Worthington 12337	NovaSeq2	9,987,939	6,628,971	57	1,384,336	191	1
Neptunia plena (L.) Benth.	R. E. Jackson V-35	NovaSeq1	14,970,785	13,958,980	57	8,241,292	325	2
Neptunia plena (L.) Benth.	R. Gonto 06173	NovaSeq2	6,053,242	4,980,126	130	2,171,079	329	0
Neptunia plena (L.) Benth.	T. R. van Devender 92–1100	NovaSeq1	15,771,863	14,860,722	57	7,906,252	330	1
Neptunia proxima A.R.Bean	K. R. McDonald 17070	MiSeq	2,757,869	2,467,769	57	3,889,577	327	0
Neptunia proxima A.R.Bean	K. R. McDonald 18188	MiSeq	2,939,533	2,611,251	57	4,216,471	326	3
Neptunia pubescens Benth.	A. N. Egan 4276	NovaSeq2	6,158,217	5,347,004	130	7,060,489	338	0
Neptunia pubescens Benth.	E. P. Killip 41174	NovaSeq2	7,875,197	4,885,746	130	1,955,754	319	2
Neptunia pubescens Benth.	Gillis 8073	NovaSeq2	8,351,872	7,431,952	57	1,352,184	297	1
Neptunia pubescens Benth.	G. Seijo 2411	NovaSeq1	12,424,997	11,636,376	57	5,882,237	328	0
Neptunia pubescens Benth.	J.d.J. Jimenez 4081	NovaSeq2	6,862,161	5,553,579	57	2,987,251	173	1
Neptunia pubescens Benth.	J. Hudson 941	NovaSeq2	5,057,220	2,664,545	130	919,570	311	0
Neptunia pubescens Benth.	J. J. Schenk 2627	NovaSeq1	23,761,488	22,332,013	57	14,355,389	335	0
Neptunia pubescens Benth.	J. J. Schenk 2627	NovaSeq1	12,967,165	12,181,620	57	5,025,183	331	0
Neptunia pubescens Benth.	L. E. Becklund 46	NovaSeq1	19,451,013	18,194,758	57	9,315,687	328	0
Neptunia pubescens Benth.	L. E. Becklund 53	NovaSeq1	20,294,247	19,122,522	57	10,632,338	333	1
Neptunia pubescens Benth.	L. E. Becklund 54	NovaSeq1	13,621,500 (363,714,688)	12,926,924	57	6,756,192	335	0
Neptunia pubescens Benth.	L. E. Becklund 65	NovaSeq1	23,293,149	21,933,133	57	9,467,622	341	0
Neptunia pubescens Benth.	L. E. Becklund 66	NovaSeq1	16,276,174	15,335,663	57	5,596,010	332	0
Neptunia pubescens Benth.	O. Ahumada 1743	NovaSeq2	4,421,676	4,030,968	57	3,690,340	175	1
Neptunia pubescens var. *microcarpa* (Rose) Windler	L. E. Becklund 57	NovaSeq1	20,090,419	19,019,035	57	10,390,955	337	1
Neptunia pubescens var. *microcarpa* (Rose) Windler	J. Rzedowski 38840	NovaSeq1	12,082,308	11,202,885	57	4,149,091	314	1
Neptunia scutata A.R.Bean	D. L. Lewis 2661	NovaSeq2	6,353,442	5,542,043	130	8,976,318	343	0
Neptunia scutata A.R.Bean	J. Egan 2995	NovaSeq2	8,105,952	6,848,804	130	11,448,163	333	0
Neptunia scutata A.R.Bean	P. R. Foulkes 421	NovaSeq2	5,231,145	4,784,881	57	6,261,747	256	4
Neptunia scutata A.R.Bean	R. W. Purdie 4866	NovaSeq2	7,897,379	6,370,204	130	10,547,518	329	1

(*Continued*)

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TABLE S2. (CONTINUED).

Species	Collector and number	Sequencing run	Input read pairs	Trimmomatic read pairs	Trimmomatic clip length	Hybpiper reads mapped	Genes with length ≥ 50%	Instances removed with PhylteR
Neptunia tactilis A.R.Bean	J. R. Clarkson 6043	NovaSeq2	6,808,584	5,679,683	130	8,892,597	335	2
Neptunia tactilis A.R.Bean	J. R. Manocochie 2168	NovaSeq2	6,334,374	4,797,609	130	7,792,082	327	1
Neptunia triquetra Benth.	L.G. J. van der Maesen 3867	NovaSeq2	5,368,807	4,867,307	57	6,704,531	280	1
Neptunia valida A.R.Bean	A. Roberts APR1079	NovaSeq2	7,827,816	6,646,304	130	10,883,580	337	2
Neptunia valida A.R.Bean	R. W. Purdie 3309	NovaSeq2	7,585,358	5,873,289	130	2,274,846	327	0
Neptunia windleriana Santos-Silva & Mansano	R. M. Harley 19075	NovaSeq2	8,084,192	5,464,829	130	2,406,468	330	1
Neptunia xanthonema A.R.Bean	A. V. Slee 3332	NovaSeq2	17,726,247	12,973,302	130	5,370,544	345	2
Neptunia xanthonema A.R.Bean	R. W. Purdie 2931	NovaSeq2	7,609,223	5,519,425	130	1,982,684	331	2
Parkinsonia aculeata L.	L. E. Becklund 56	NovaSeq1	24,633,788	23,258,552	57	13,289,206	332	1
Prosopidastrum mexicanum (Dressler) Burkart	R. Moran 17061	NovaSeq1	10,740,976	10,052,688	57	3,659,494	302	3

TABLE S3. The corrected Akaike information criterion (AICc) values of the ancestral character state transition models, by character state.

State Model	AICc	Delta-AICc
Habitat		
Equal Rates	19.71	1.17
All Rates Different	18.54	0
Nectary		
Equal Rates	45.61	0
All Rates Different	47.84	2.23
Anther Gland		
Equal Rates	46.17	0
All Rates Different	48.27	2.10
Seed Number		
Equal Rates	17.24	0
All Rates Different	19.15	1.91

APPENDIX 2. Sampled taxa and number of loci used in BEAST analyses. Order of the data are sample name and abbreviated collection number (corrected taxa name, if applicable), number of loci. The 14 Angiosperms353 loci included in the BEAST analyses were 4796, 5335, 5339, 5398, 5404, 5454, 5634, 5933, 6098, 6258, 6450, 6462, 6488, and 7013. Sequence data for *Desmanthus virgatus* was missing in loci 5398, 5404, 6098, and 6462.

Ingroup: *Neptunia amplexicaulis* F5856, 14; *Neptunia gracilis* var. *glandulosa* BW722 (*Neptunia dimorphantha*), 14; *Neptunia heliophila* JRH2177 (*Neptunia gracilis* s.l.), 14; *Neptunia hispida* ML7779, 14; *Neptunia insignis* IGC2139, 14; *Neptunia javanica* OEW20, 14; *Neptunia longipila* SvL5099, 14; *Neptunia lutea* DJR4801, 14; *Neptunia major* HD8836, 14; *Neptunia monosperma* DLL1068, 14; *Neptunia oleracea* BR95, 14; *Neptunia paucijuga* NJC662, 14; *Neptunia plena* ARN4253, 14; *Neptunia proxima* KRM17070 (*Neptunia proxima* a), 14; *Neptunia proxima* KRM18188 (*Neptunia proxima* b), 14; *Neptunia pubescens* LEB65, 14; *Neptunia scutata* JE2995, 14; *Neptunia tactilis* JRC6043, 14; *Neptunia triquetra* LGJvdM3867, 14; *Neptunia valida* ARAPR1079, 14; *Neptunia xanthonema* AVS3332, 14.

Outgroup: *Acaciella angustissima* LEB61, 14; *Albizia julibrissin* LEB67, 14; *Calliandra eriophylla* LEB73, 14; *Desmanthus velutinus* LEB58, 14; *Desmanthus virgatus* LEB48, 10; *Dichrostachys cinerea* ERR7618736, 14; *Lemurodendron capuronii* EJMK435, 14; *Mimosa aculeaticarpa* LEB55, 14; *Neltuma glandulosa* LEB49, 14; *Parkinsonia aculeata* LEB56, 14; *Prosopidastrum mexicanum* RM17061, 14.

APPENDIX 3. Ancestral states of habitat (aquatic, terrestrial), petiole nectary (present, absent), anther gland (present, absent), and seeds per fruit (single, multiple) for each species.

Ingroup: *Neptunia amplexicaulis*, Terrestrial, Absent, Absent, Multi-seeded; *Neptunia dimorphantha*, Terrestrial, Present, Present, Multi-seeded; *Neptunia gracilis* sl, Terrestrial, Absent, Present, Multi-seeded; *Neptunia hispida*, Terrestrial, Absent, Present, Multi-seeded; *Neptunia insignis*, Terrestrial, Absent, Absent, Multi-seeded; *Neptunia javanica*, Terrestrial, Present, Absent, Multi-seeded; *Neptunia longipila*, Terrestrial, Present, Absent, Single-seeded; *Neptunia lutea*, Terrestrial, Absent, Present, Multi-seeded; *Neptunia major*, Terrestrial, Present, Absent, Multi-seeded; *Neptunia monosperma*, Terrestrial, Present, Absent, Single-seeded; *Neptunia oleracea*, Aquatic, Absent, Absent, Multi-seeded; *Neptunia paucijuga*, Terrestrial, Present, Present, Multi-seeded; *Neptunia plena*, Aquatic, Present, Present, Multi-seeded; *Neptunia proxima* a, Terrestrial, Absent, Absent, Multi-seeded; *Neptunia proxima* b, Terrestrial, Absent, Absent, Multi-seeded; *Neptunia pubescens*, Terrestrial, Absent, Present, Multi-seeded; *Neptunia scutata*, Terrestrial, Present, Absent, Single-seeded; *Neptunia tactilis*, Terrestrial, Absent, Present, Multi-seeded; *Neptunia triquetra*, Terrestrial, Present, Present, Multi-seeded; *Neptunia valida*, Terrestrial, Absent, Present, Single-seeded; *Neptunia xanthonema*, Terrestrial, Present, Absent, Single-seeded.

Outgroup: *Acaciella angustissima*, Terrestrial, Absent, Absent, Multi-seeded; *Albizia julibrissin*, Terrestrial, Present, Absent, Multi-seeded; *Calliandra eriophylla*, Terrestrial, Absent, Absent, Multi-seeded; *Desmanthus velutinus*, Terrestrial, Present, Absent, Multi-seeded; *Desmanthus virgatus*, Terrestrial, Present, Absent, Multi-seeded; *Dichrostachys cinerea*, Terrestrial, Present, Present, Multi-seeded; *Lemurodendron capuronii*, Terrestrial, Present, Present, Multi-seeded; *Mimosa aculeaticarpa*, Terrestrial, Absent, Absent, Multi-seeded; *Neltuma glandulosa*, Terrestrial, Present, Present, Multi-seeded; *Parkinsonia aculeata*, Terrestrial, Absent, Absent, Multi-seeded; *Prosopidastrum mexicanum*, Terrestrial, Present, Present, Multi-seeded.